Efficiency in Job-Ladder Models*

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Abstract

This paper examines the efficiency of a decentralized equilibrium in a broad class of random-search job-ladder models. We show that, with job ladders, the Hosios condition no longer guarantees efficiency: the decentralized steady-state equilibrium features excessive vacancy creation. We decompose the inefficiency into an investment margin, which concerns vacancy creation given match values, and a valuation margin, which concerns the gap between private and social values of a match. Sequential auctions, together with the Hosios condition, correct the investment margin but not the valuation margin. Quantitatively, under a standard calibration, the decentralized unemployment rate is about one percentage point below the efficient level and responds too strongly to productivity shocks. We also characterize taxes that decentralize the efficient allocation.

Keywords: search and matching, job ladder, on-the-job search, Hosios condition

JEL Classification: D61, D62, E24, J62, J63, J64

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1 Introduction

In designing and evaluating macroeconomic policies in frictional labor markets, understanding the normative properties of equilibrium is essential. In a frictional environment, assessing the efficiency of a decentralized equilibrium is a nontrivial task. Since matching workers and firms is costly, the mere presence of unemployment is not necessarily a sign of inefficiency. The important question is instead whether the decentralized allocation creates the right number and types of matches, taking the underlying search frictions as given.

A natural benchmark for answering this question is the *Hosios condition*. In the standard Diamond-Mortensen-Pissarides model without on-the-job search, Hosios (1990) shows that the decentralized equilibrium is constrained efficient when the worker's bargaining power equals the elasticity of the matching function. This result has become a standard benchmark in macroeconomic applications of search-and-matching models. The Hosios condition is frequently used for calibration in the quantitative macro literature (e.g., Shimer, 2005; Faberman et al., 2022), and it is referenced even when the elasticity of the matching function and the worker's bargaining power are separately calibrated (e.g., Lise et al., 2016; Bilal et al., 2022). Various papers that extend the standard DMP model in different directions also use the Hosios result as the point of comparison (e.g., Acemoglu, 2001; Davis, 2001; Mukoyama, 2019).

In this paper, we show that the Hosios condition no longer guarantees efficiency once on-the-job search is taken into account. In fact, with on-the-job search, the decentralized steady-state equilibrium exhibits excessive vacancy creation under the Hosios condition. Excessive vacancy creation causes workers to climb the job ladder too quickly, and the decentralized equilibrium has *too many* high-quality matches compared with the efficient allocation. This finding is in contrast to a model with ex-ante job heterogeneity such as Acemoglu (2001) that has *too few* good jobs in the decentralized equilibrium.

This result is important because job-to-job transitions are central to modern analyses

of labor-market dynamics. In particular, the importance of *job ladders*, that is, the process by which workers move to better jobs through on-the-job search, is well documented. At the micro level, job ladders are important determinants of earnings growth and workers' career development.[1] At the macro level, job ladders shape the dynamics of aggregate productivity, wages, inflation, and unemployment.[2]

Despite their importance and popularity, however, the normative properties of job-ladder models are not well understood. We provide intuition for our results by decomposing inefficiency into two margins. The *investment margin* asks whether firms create the socially efficient number of vacancies, taking the value of a match as given. The *valuation margin* asks whether the private value of a match coincides with its social value. In the standard DMP model of Pissarides (1985), the Hosios condition aligns both margins. In job-ladder models, the Hosios condition no longer aligns both margins simultaneously.

On the investment margin, vacancy creation affects not only unemployed workers but also employed workers who may be poached from their current employers. When a worker is poached, the existing match is destroyed. In the decentralized equilibrium, this loss is not necessarily fully internalized by the poacher, leading to a "worker-stealing" effect, similar to the business-stealing effect in the economic growth literature. We demonstrate that imposing the sequential-auction wage-setting protocol of Dey and Flinn (2005) and Cahuc et al. (2006), together with the Hosios condition, forces the poacher to internalize the worker-stealing effect through offer matching, thereby achieving efficiency along the investment margin.

The key result of the paper is that, even when the investment margin is efficient under the Hosios condition and sequential auction, the valuation margin remains distorted in

[1] For example, Topel and Ward (1992) attribute about one-third of earnings growth for young workers to job-to-job transitions. More recently, Hahn et al. (2021) show that workers gain both wages and hours, on average, through job-to-job transitions.

[2] For example, job ladders affect worker reallocation and allocative efficiency (Barlevy, 2002), wage dispersion (Postel-Vinay and Robin, 2002; Hornstein et al., 2011), inflation (Moscarini and Postel-Vinay, 2023; Faccini and Melosi, 2023; Birinci et al., 2024), wage rigidity (Fukui, 2020), unemployment dynamics (Moscarini and Postel-Vinay, 2018; Faberman et al., 2022), and aggregate productivity dynamics (Mukoyama, 2013).

the presence of on-the-job search. The private and social values of a match differ because of how they value on-the-job search relative to off-the-job search. For the planner, the key consideration is how much on-the-job search congests the market relative to off-the-job search. For private agents, the key consideration is how much surplus from on-the-job search, relative to off-the-job search, is appropriated by other parties through bargaining. Without on-the-job search, the Hosios condition ensures that these considerations coincide: the extent to which off-the-job search congests the market equals the extent to which surplus from off-the-job search is appropriated by the newly hiring firm. We show that the presence of on-the-job search leads them to diverge in a way that causes matches to be overvalued in the decentralized equilibrium.

The intuition is clearest in the case where on-the-job search efficiency is the same as off-the-job search efficiency. In that case, from the planner's perspective, on-the-job search congests the market in the same way as off-the-job search. By contrast, from the private agent's perspective, the surplus appropriated from on-the-job search is smaller, because the gains from moving to a better job are smaller than the gains from leaving unemployment. As a result, private agents place a higher value on on-the-job search relative to off-the-job search than the planner does, leading to the overvaluation of jobs in equilibrium. This overvaluation of jobs, in turn, leads to excessive vacancy creation.

More generally, our main theoretical result proves that the decentralized equilibrium features excessive vacancy creation under the Hosios condition for a wide range of wage-setting protocols that span from no offer matching to complete offer matching (i.e., sequential auction). As explained above, sequential auction implies that the investment margin is efficient but jobs are overvalued along the valuation margin, resulting in excessive vacancy creation. Imperfect offer matching further exacerbates the over-creation of jobs because poachers fail to internalize the business-stealing effect along the investment margin. As a result, vacancy creation remains excessive regardless of the extent of offer matching.

We then show that the efficient allocation can be implemented with a combination of a vacancy tax, which corrects for the investment margin, and an employment tax, which corrects for the valuation margin. Under the Hosios condition, the vacancy tax is zero when wages are set by sequential auction and positive otherwise. The employment tax is always positive and increases with the job-ladder rank, which implies that the overvaluation of jobs is particularly pronounced at the top of the job ladder. Intuitively, the surplus appropriated by the newly hiring firm is particularly small near the top of the job ladder, while the congestion effect remains the same. Imperfect appropriation therefore fails to offset the congestion effect especially at the top of the job ladder.

We then quantify the discrepancy between the decentralized equilibrium and the efficient allocation. We find that the discrepancy is quantitatively significant under a standard calibration from the literature that imposes the Hosios condition and sequential auctions. The decentralized equilibrium has an unemployment rate 0.9 percentage points lower than that in the efficient allocation. Finally, we show that equilibrium unemployment is excessively volatile in response to a positive productivity shock. This result is due to the fact that the overvaluation of jobs is further exacerbated when the labor market is tight.

Our paper is related to several strands of the literature. The most directly related are Gautier et al. (2010) and Cai (2020), who study efficiency in the presence of on-the-job search. Their analyses use a framework that is not typically used in macroeconomics and also impose several parametric restrictions, such as equal on- and off-the-job search efficiency, equating the discount rate with the population growth rate, and constant wages during the match. Our analysis builds on a widely used DMP framework with a general wage-setting protocol that nests sequential auctions (Dey and Flinn, 2005; Cahuc et al., 2006), and makes no such restrictions on the discount rate or search efficiency. Beyond these differences in environment, our main contribution relative to these papers is the explicit decomposition into investment and valuation margins.

The second strand of related literature is the analysis of the DMP model with heterogeneous jobs. Papers such as Acemoglu (2001), Davis (2001), and Mukoyama (2019) explicitly consider efficiency in DMP models with heterogeneous jobs and no on-the-job search. These papers do not deal with job-to-job transitions.

Finally, previous studies have examined the efficiency of equilibrium in directed-search settings. For example, Menzio and Shi (2011) establish the efficiency of directed-search equilibrium in a model with on-the-job search. Our paper considers random-search models, where different types of workers interact in the same labor market. In line with Menzio and Shi's (2011) result, we show that our economy is efficient once the matching market is segmented by the job ladder. Given that reality lies somewhere between the random and the segmented labor market, our results provide an important benchmark for the polar cases.

This paper is organized as follows. Section 2 sets up the terminology we use in our analysis by reviewing the intuition of the Hosios (1990) condition in a model without on-the-job search. Section 3 describes our model and compares the efficient allocation and the equilibrium outcome. Section 4 derives the main analytical results, including the decentralization of the efficient allocation. Section 5 presents the quantitative analysis. Section 6 concludes.

2 Revisiting the basic intuition of the Hosios condition

Before considering our baseline model, let us set up the textbook Pissarides (1985) model without on-the-job search to develop basic intuition. The economy is populated by a unit continuum of infinitely lived, risk-neutral workers with discount rate $r > 0$. Workers are either employed in a one-worker, one-firm match that produces $z > 0$, or unemployed with flow home-production value $h \in [0, z)$. Firms post vacancies at flow cost $\kappa > 0$ under free entry. Matching is random and governed by $M(u, v)$, which is strictly increasing and

concave and exhibits constant returns to scale. There is no on-the-job search, and only unemployed workers search. Job-finding and vacancy-filling rates are

$$p(\theta) \equiv M(1,\theta) \quad \text{and} \quad q(\theta) \equiv M\left(\frac{1}{\theta},1\right),$$

where $\theta \equiv v/u$ is market tightness and $p(\theta) = \theta q(\theta)$.

We assume that separation is random with Poisson rate $\sigma > 0$. In decentralized equilibrium, wages are determined according to the generalized Nash bargaining solution with the worker's bargaining power $\gamma \in [0,1)$.

The details of the social planner's problem and the decentralized equilibrium are analyzed in Appendix A. It shows that the Hosios condition (Hosios, 1990)

$$\gamma = \eta(\theta), \tag{1}$$

where $\eta(\theta) \equiv -\theta q'(\theta)/q(\theta)$ is the elasticity of the matching function, ensures the constrained efficiency of the market outcome. That is, under condition (1), the solution to the social planner's problem coincides with the decentralized equilibrium. Intuitively, the Hosios condition can be understood as the condition where two inefficiencies exactly cancel each other out.

Let us review the intuition more closely. For ease of exposition, we focus on the steady state. First, consider the optimization problem for vacancy posting. This exercise amounts to comparing

$$\kappa = (1-\eta(\theta))q(\theta)\mu \tag{2}$$

for the social planner's problem, where μ is the social value of moving one additional unemployed worker into employment, and

$$\kappa = (1-\gamma)q(\theta)S, \tag{3}$$

where S is the surplus of a match between a worker and a firm. Here, suppose $\mu = S$ holds. Even in that situation, two inefficiencies arise when comparing equations (2) and (3). We refer to this margin as the "investment margin" because it relates to the firm's vacancy creation as an investment. The first inefficiency is the hold-up problem. Matches are formed because of the firms' active investment (vacancy posting), and workers incur no costs. Thus, all returns from the match should be paid to the firms to ensure an efficient level of investment. However, at the time the worker and the firm engage in Nash bargaining, the investment costs are already sunk. As a result, the firm can collect only a $(1-\gamma)$ share of the surplus (the hold-up problem) on the right-hand side of (3). This inefficiency, resulting from firms' imperfect appropriation of surplus, leads to too few vacancies compared to the socially desirable level. When γ (the worker's bargaining power) is large, the inefficiency is large.

The second inefficiency is the congestion externality. Posting a vacancy congests the matching process for other firms, which discourages them from posting vacancies. Privately, the posting firm does not internalize that congestion, which is a force toward too many vacancies. In equation (2), this congestion is captured by the factor $(1-\eta(\theta))$: the social return to a vacancy is the marginal meeting rate $M_2(u,v) = (1-\eta(\theta))q(\theta)$, rather than the private meeting rate $q(\theta)$.

Now, consider how the inefficiency shows up in the calculation of the social and private values of the match. We refer to this margin as the "valuation margin." This exercise compares

$$(r+\sigma)\mu = z - h - (p(\theta)\mu - \kappa\theta) \tag{4}$$

and

$$(r+\sigma)S = z - h - p(\theta)\gamma S. \tag{5}$$

In equation (4), the (flow) social value of the match is the sum of the current surplus $z-h$ and the opportunity cost of keeping a worker employed. The opportunity cost is that, by

keeping the worker unemployed, she could have generated a new match with probability $p(\theta)$. However, this calculation does not take into account that increasing u changes θ if v is kept constant. For an infinitesimal increase du in the measure of unemployed workers, holding θ constant requires $dv = \theta\, du$, since $d\theta = (dv - \theta\, du)/u = 0$. The additional vacancies cost $\kappa\theta\, du$, so the cost per unit increase in unemployment is $\kappa\theta$, which is subtracted from the gain from matching. Using the free-entry condition (2), this cost can be written as $\kappa\theta = p(\theta)(1 - \eta(\theta))\mu$.

In the decentralized equilibrium equation (5), the second term on the right-hand side, $p(\theta)\gamma S$, represents the worker's opportunity cost from working (not searching). The full opportunity cost is $p(\theta)S$, but here, the fraction $p(\theta)(1-\gamma)S$ is unaccounted for because the worker only receives a γ fraction of the surplus. The private value corresponds to $p(\theta)S$, and the externality term is $\kappa\theta$ (it is the amount of resources required to "undo" the externality). Under the Hosios condition, this inefficiency is offset by the inefficiency that the worker only recognizes the γ fraction of the opportunity cost (the surplus it could have created by being unemployed).

As we can see, the Hosios condition achieves "offsetting one inefficiency by another inefficiency" on the firm side as well as the worker side. It can "kill two birds with one stone" because both inefficiencies are symmetric: the firms and the unemployed workers create similar externalities, and they are on the opposite sides of the bargaining. Moreover, the direction of the total inefficiencies is aligned when the Hosios condition does not hold: when γ is too small, S is too large in the valuation margin. These two imply $(1-\gamma)S$ is too large in the investment margin. Thus, θ is too large in equilibrium.

It is useful to note that, in both margins, the nature of the inefficiencies is different even though they can exactly offset each other via the Hosios condition. In particular, the search externality is imposed on *other firms*, whereas the imperfect appropriation is about the *own match*.[3]

[3]Also note that the valuation margin does not show up in a two-period model where the first-period search creates the second-period match, because the inefficiency stems from (i) the externality imposed on

3 The model with on-the-job search

In this section, we present our main model. The model now features on-the-job search.

3.1 Decentralized equilibrium

The economy is populated by a unit mass of workers. All workers have the following linear preferences:

$$\int_0^\infty e^{-rt} c_t dt,$$

where $r > 0$ is, once again, the discount rate, and c_t is the consumption at t. Workers receive wages (and consume them) while working and enjoy h units of consumption (home production) while unemployed.

Here, the new assumption is that workers search both on and off the job. That is, matches are heterogeneous, and an employed worker may meet with a new job (new match). When the new job is better, the worker moves. Thus, over time, a worker may climb the job ladder.

Firms create vacancies at per-vacancy cost $\kappa > 0$. When firms meet with workers, they draw permanent match quality, $z \in [0, \infty)$, with pdf $g(z)$ and cdf $G(z)$. We assume the match quality has a finite mean. Firms with match quality z produce $A_t z$ units of output per unit time, where A_t is the aggregate productivity at time t.

The economy is subject to matching frictions. We normalize the search efficiency of unemployed workers to one. Employed workers have a search intensity of $\zeta \in [0, 1]$. The total efficiency of the search on the worker side is

$$x_t \equiv u_t + \zeta(1 - u_t).$$

the unemployed workers' search in the second period and (ii) the failure to take the opportunity cost of the second-period search properly into account.

We assume the matching is random. Let

$$f_t^u \equiv \frac{u_t}{x_t}, \quad f_t(z) \equiv \frac{\zeta n_t(z)}{x_t} \tag{6}$$

be the probability (conditional on the meeting) of a vacancy encountering unemployed workers and the probability density (conditional on the meeting) of a vacancy encountering employed workers with match quality z, respectively. Here, $n_t(z)$ denotes the measure of workers employed at z.

Given the number of vacancies created (v_t), the number of matches in the economy is given by the matching function $M(x_t, v_t)$, where we assume this function is increasing in both terms and exhibits constant returns to scale. We denote

$$\theta_t \equiv \frac{v_t}{x_t}$$

as the labor market tightness. The Poisson rates at which an unemployed worker meets a vacancy and a vacancy meets a worker are $p(\theta_t) \equiv M(1, \theta_t)$ and $q(\theta_t) \equiv M(1/\theta_t, 1)$, respectively. The meeting rate of workers employed in z is $\zeta p(\theta_t)$. The match separates exogenously at rate $\sigma > 0$ and potentially endogenously.

We let U_t denote a worker's value of being unemployed, $W_t(z, \bar{O})$ denote the value of an employed worker with match quality z and the worker's outside option $\bar{O}$, $J_t(z, \bar{O})$ denote the value of a firm with match quality z and the worker's outside option $\bar{O}$, and V_t denote the value of a vacancy. Throughout, $\bar{O}$ denotes the worker's outside option *in excess of* the value of unemployment, so that $\bar{O} = 0$ corresponds to the value of unemployment. Define the joint match surplus to be

$$S_t(z) \equiv W_t(z, \bar{O}) + J_t(z, \bar{O}) - U_t - V_t, \tag{7}$$

where $S_t(z)$ is independent of the worker's outside option, which we confirm in our equi-

librium construction below.

As is common in the literature, we look for a monotone Markov equilibrium, in which (i) the firm's and the worker's strategies only depend on the payoff-relevant state variables $(z, \bar{O})$ as well as calendar time t, and (ii) the surplus function $S_t(z)$ is strictly increasing in z.[4]

The wages are determined by Nash bargaining, where the worker's outside option may depend on the history of past meetings.[5] When unemployed workers decide to form a match, they engage in Nash bargaining with the worker's bargaining weight $\gamma \in [0, 1)$. The outside option for unemployed workers is the unemployment value U_t.

When employed workers at firm z with excess outside option $\bar{O}$ meet with poachers z', the following takes place. If $S_t(z) \geq S_t(z')$, workers stay and bargain with excess outside option $\max\{\bar{O}, \omega S_t(z')\}$, where $\omega \in [0, 1]$ is a parameter governing the degree of offer matching and $S_t(z')$ is the joint match surplus of job z', which we define below. When $S_t(z) < S_t(z')$, workers are poached, and the excess outside option of workers is given by $\omega S_t(z)$. Between meetings, the excess outside option is continuously revalued using the current surplus of the offer underlying it, with unemployment as the fallback. The offer-matching parameter ω is sufficiently flexible to nest many existing wage-setting protocols. Two important special cases are the following:

1. Nash bargaining with no commitment (e.g., McCrary, 2022; Gottfries, 2026): $\omega = 0$;

[4] There exist non-Markovian equilibria in which workers form future wage expectations based on current wages, in which case wages play an allocative role (see, e.g., Coles, 2001). There may also exist non-monotone equilibria in which the surplus is decreasing in productivity. See Gottfries (2026) for such an example.

[5] Shimer (2006) observes that in a class of search and matching models with on-the-job search, the feasible payoffs become nonconvex and the Nash bargaining solution becomes inapplicable. His logic involves the mechanism by which the current wage decreases the future probability of worker poaching, and thus applies only to models where the wage from the current employer does not change after a meeting with another (potential) employer. In our setting, wages are continuously renegotiated, and continuation play depends only on match quality and the worker's outside option, not on the current wage (Markov restriction). Under these assumptions, the issue raised by Shimer (2006) does not arise. Other papers we cite below, such as Cahuc et al. (2006) and Borovičková and Shimer (2024), follow the same approach. We could have also assumed that renegotiation takes place only upon meeting with the poaching firm, which does not make any difference in the steady state.

2. Sequential auction (e.g., Dey and Flinn, 2005; Cahuc et al., 2006): $\omega = 1$.[6]

We will highlight these two cases later on.

The value function of the unemployed is

$$rU_t = h + p(\theta_t) \int g(z) \max\{W_t(z,0) - U_t, 0\} dz + \dot{U}_t, \tag{8}$$

where the "dot notation" represents the time derivative: $\dot{U}_t \equiv \partial U_t / \partial t$. The value function of workers employed in firm z with an excess outside option $\bar{O} \in [0, \omega S_t(z)]$ is given by

$$\begin{aligned} rW_t(z,\bar{O}) = w_t(z,\bar{O}) &+ \zeta p(\theta_t) \int_z^\infty g(z') \left(W_t(z', \omega S_t(z)) - W_t(z,\bar{O})\right) dz' \\ &+ \zeta p(\theta_t) \int_0^z g(z') \left(W_t(z, \max\{\omega S_t(z'), \bar{O}\}) - W_t(z,\bar{O})\right) dz' \\ &+ \sigma(U_t - W_t(z,\bar{O})) + \dot{W}_t(z,\bar{O}). \end{aligned} \tag{9}$$

The value of the filled job with productivity z and outside option $\bar{O} \in [0, \omega S_t(z)]$ is given by

$$\begin{aligned} rJ_t(z,\bar{O}) = A_t z - w_t(z,\bar{O}) &+ \zeta p(\theta_t) \int_z^\infty g(z') \left(V_t - J_t(z,\bar{O})\right) dz' \\ &+ \zeta p(\theta_t) \int_0^z g(z') \left(J_t(z, \max\{\omega S_t(z'), \bar{O}\}) - J_t(z,\bar{O})\right) dz' \\ &+ \sigma \left(V_t - J_t(z,\bar{O})\right) + \dot{J}_t(z,\bar{O}). \end{aligned} \tag{10}$$

Here, $\dot{W}_t(z,\bar{O})$ and $\dot{J}_t(z,\bar{O})$ denote total time derivatives including the revaluation of the excess outside option between meetings. Given the above bargaining protocol, the outside option $\bar{O}$ cannot exceed $\omega S_t(z)$ for a match with productivity z along the equilibrium

[6] A subtle point is that here we assume the wage renegotiation occurs continuously, as in the standard DMP model, whereas in Dey and Flinn (2005) and Cahuc et al. (2006), a negotiation occurs only when a new meeting happens. This difference does not result in different outcomes in the steady state.

path. Thus, we ignore such a possibility. The value of a vacancy is given by

$$\begin{aligned} rV_t = -\kappa + f_t^u q(\theta_t) \int_0^\infty g(z) \max\{J_t(z,0) - V_t, 0\} dz \\ + q(\theta_t) \int_0^\infty f_t(z) \int_z^\infty g(z') \max\{J_t(z', \omega S_t(z)) - V_t, 0\} dz' dz + \dot{V}_t, \end{aligned} \tag{11}$$

where $\kappa > 0$ is the vacancy-posting cost. We assume free entry into vacancy posting. This assumption implies a zero value of vacancy posting:

$$V_t = 0. \tag{12}$$

The Nash bargaining with the worker's outside option $\bar{O}$ solves

$$\max_{w_t(z,\bar{O})} \left(W_t(z,\bar{O}) - (U_t + \bar{O})\right)^\gamma \left(J_t(z,\bar{O}) - V_t\right)^{1-\gamma}.$$

After imposing the free-entry condition (12), the solution to the bargaining problem yields

$$W_t(z,\bar{O}) = U_t + \bar{O} + \gamma(S_t(z) - \bar{O}) \tag{13}$$

and

$$J_t(z,\bar{O}) = (1-\gamma)(S_t(z) - \bar{O}). \tag{14}$$

Plugging (8), (9), and (10) into (7) and using (13), we obtain the recursive expression for the job-match surplus:

$$\begin{aligned} (r+\sigma)S_t(z) = A_t z - h + \zeta p(\theta_t) \int_z^\infty g(z') \left(\omega S_t(z) + \gamma \left[S_t(z') - \omega S_t(z)\right] - S_t(z)\right) dz' \\ - p(\theta_t) \int_{\underline{z}_t}^\infty g(z') \gamma S_t(z') dz' + \dot{S}_t(z). \end{aligned} \tag{15}$$

The reservation match quality $\underline{z}_t$, above which the match is formed, satisfies

$$S_t(\underline{z}_t) = 0. \tag{16}$$

We assume that the lower bound of the distribution of match quality is low enough so that the reservation match quality $\underline{z}_t$ is interior to the support.[7] The expression (15) confirms our original presumption that the joint match surplus is independent of the worker's outside option $\bar{O}$.

After imposing the free-entry condition (12) and also (14) in (11), we obtain

$$\kappa = (1-\gamma)q(\theta_t)\left[f_t^u \int_{\underline{z}_t}^{\infty} g(z)S_t(z)dz + \int_0^{\infty}\int_z^{\infty} f_t(z)g(z')\left[S_t(z') - \omega S_t(z)\right]dz'dz\right] \tag{17}$$

For active matches, $z > \underline{z}_t$, the employment density satisfies the Kolmogorov forward equation

$$\dot{n}_t(z) = p(\theta_t)g(z)\left[u_t + \zeta\int_{\underline{z}_t}^{z} n_t(z')dz'\right] - \left[\sigma + \zeta p(\theta_t)(1-G(z))\right]n_t(z). \tag{18}$$

The first term is an inflow from unemployment and hires from lower rungs of the job ladder. The second term is an outflow due to exogenous separation (with Poisson rate σ) and poaching from the higher rungs of the job ladder. Matches below the reservation quality are terminated immediately, so $n_t(z) = 0$ for $z < \underline{z}_t$. Because our model abstracts from idiosyncratic match quality shocks, endogenous separation never occurs in the steady state, but it can occur in the transitional dynamics.

The unemployment rate is determined by

$$u_t = 1 - \int_{\underline{z}_t}^{\infty} n_t(z)dz. \tag{19}$$

[7] In Appendix H, we study a case where the reservation match quality margin is absent and prove the same set of results as in the baseline model.

Together with the cutoff destruction rule, this identity accounts for the increase in unemployment when existing matches become unviable.

We are now ready to define equilibrium.

Definition 1 *Given* $\{n_0(z), u_0\}$*, a monotone Markov equilibrium consists of a sequence of the joint match surplus function* $\{S_t(z)\}$*, the employment distribution across the job ladder,* $\{n_t(z), f_t(z), f_t^u\}$*, and the unemployment rate,* $\{u_t\}$*, market tightness,* $\{\theta_t\}$*, and reservation match quality* $\{\underline{z}_t\}$ *such that* (6)*,* (15)*,* (16)*,* (17)*,* (18)*, and* (19) *hold. The steady-state equilibrium is the one where all variables are constant over time.*

3.2 Social planner's problem

The social planner directly controls the offer-acceptance decisions of all workers and thus controls all worker flows (subject to matching frictions), as well as the creation of vacancies. Let $\mathbb{I}_t^{UE}(z)$ be an indicator function that takes the value 1 if an unemployed worker who meets a job with match quality z accepts the offer. Likewise, let $\mathbb{I}_t^{EE}(z, z')$ be an indicator function that takes the value 1 if an employed worker with match quality z who meets a job with match quality z' accepts the offer. Finally, $\varsigma_t(z) \geq 0$ denotes the density of the endogenous separation flow.

The social planner's problem is to choose $\{\theta_t, n_t(z), \mathbb{I}_t^{UE}(z), \mathbb{I}_t^{EE}(z, z'), \varsigma_t(z)\}$ to maximize

$$\int_0^\infty e^{-rt} \left[\int_0^\infty A_t z n_t(z) dz + h \left(1 - \int_0^\infty n_t(z) dz \right) - \kappa \theta_t \left(1 - \int_0^\infty n_t(z) dz + \int_0^\infty \zeta n_t(z) dz \right) \right] dt$$

subject to

$$\begin{aligned} \dot{n}_t(z) = & \left(1 - \int n_t(z') dz' \right) p(\theta_t) g(z) \mathbb{I}_t^{UE}(z) + \int_0^\infty p(\theta_t) g(z) \mathbb{I}_t^{EE}(z', z) \zeta n_t(z') dz' \\ & - \int_0^\infty p(\theta_t) g(z') \mathbb{I}_t^{EE}(z, z') \zeta n_t(z) dz' - \sigma n_t(z) - \varsigma_t(z). \end{aligned} \tag{20}$$

The first term in the square brackets in the objective function is the production by active matches, the second term is the home production, and the third term is the vacancy-posting cost κv_t. For the constraint, the change in the number of matches with the match quality z is due to (i) matches created with workers moving from U to E plus (ii) E to E movements from another z' to z minus (iii) E to E movements out of z (to another z') and separation.

Let $\mu_t(z)$ be the costate variable that represents the shadow value of the constraint (20). Thus, $\mu_t(z)$ is the shadow value of creating one additional match with match quality z. For this reason, we refer to $\mu_t(z)$ as the match surplus from the planner's perspective.

The optimality conditions for $\{\mathbb{I}_t^{UE}(z), \mathbb{I}_t^{EE}(z,z')\}$ are

$$\mathbb{I}_t^{UE}(z) = \begin{cases} 1 & \mu_t(z) > 0 \\ 0 & \mu_t(z) \leq 0 \end{cases}, \qquad \mathbb{I}_t^{EE}(z,z') = \begin{cases} 1 & \mu_t(z') > \mu_t(z) \\ 0 & \mu_t(z') \leq \mu_t(z). \end{cases}$$

The pointwise optimality condition for the nonnegative endogenous separation flow $\varsigma_t(z)$ implies $\mu_t(z) \geq 0$ for all z with $n_t(z) > 0$.

The first-order optimality condition on $n_t(z)$ is

$$\begin{aligned}(r+\sigma)\mu_t(z) = A_t z - h - \int_{\underline{z}_t}^{\infty} p(\theta_t) g(z') \mu_t(z') dz' + \zeta p(\theta_t) \int_z^{\infty} g(z')(\mu_t(z') - \mu_t(z)) dz' \\ + \kappa\theta_t(1-\zeta) + \dot{\mu}_t(z),\end{aligned} \tag{21}$$

where we have already imposed the fact that $\mu_t(z)$ is increasing in z. The reservation match quality $\underline{z}_t$, above which the planner forms a match, satisfies

$$\mu_t(\underline{z}_t) = 0. \tag{22}$$

We can rewrite the first-order optimality condition for θ_t as

$$\kappa = (1-\eta(\theta_t))q(\theta_t)\left[f_t^u \int_{\underline{z}_t}^{\infty} g(z')\mu_t(z')dz' + \int_{\underline{z}_t}^{\infty}\int_{z}^{\infty} f_t(z)g(z')\left(\mu_t(z') - \mu_t(z)\right)dz'dz\right], \tag{23}$$

where f_t^u and $f_t(z)$ are defined as in (6) and $\eta(\theta) \equiv -\theta q'(\theta)/q(\theta)$. The left-hand side is the cost of posting one additional vacancy, and the right-hand side is the marginal increase in the matching probability $p'(\theta_t) = (1-\eta(\theta_t))q(\theta_t)$ times the value of the match the vacancy creates. With frequency f_t^u, the vacancy meets an unemployed worker and generates value $\mu_t(z')$ with probability (density) $g(z')$ (when $\mu_t(z') > 0$). With frequency $f_t(z)$, it meets with an employed worker with quality z and draws the new match quality z' with density $g(z')$. When $\mu_t(z') > \mu_t(z)$, the worker moves to the new job, and the social value $\mu_t(z') - \mu_t(z)$ is generated.

3.3 Efficiency

Following the same steps as in Section 2, we organize the source of the discrepancy between the planner's problem and the decentralized equilibrium into two margins: the investment margin and the valuation margin.

3.3.1 Leading example under the Hosios condition and sequential auction

We will lead our discussion with an economy with the Hosios condition ($\gamma = \eta(\theta)$) and sequential auction ($\omega = 1$). Sequential auction is a particularly popular assumption in the literature, and moreover, this assumption simplifies the analysis and provides a useful stepping stone for the subsequent analysis. We highlight that even with these two conditions, the decentralized equilibrium is inefficient in an unambiguous direction. We later generalize this result to the case with arbitrary $\omega \in [0,1]$. For simplicity, we focus on the steady state and drop time subscripts.

We start by comparing the free-entry condition in the decentralized equilibrium and

the planner's problem. In the decentralized equilibrium with the Hosios condition and sequential auction, the free-entry condition (17) becomes:

$$\kappa = (1-\eta(\theta))q(\theta)\left[f^u\int_{\underline{z}}^{\infty} g(z)S(z)dz + \int_0^{\infty}\int_z^{\infty} f(z)g(z')\left[S(z')-S(z)\right]dz'dz\right].$$

In the planner's solution, the analogous free-entry condition is (23):

$$\kappa = (1-\eta(\theta))q(\theta)\left[f^u\int_{\underline{z}}^{\infty} g(z')\mu(z')dz' + \int_{\underline{z}}^{\infty}\int_z^{\infty} f(z)g(z')\left(\mu(z')-\mu(z)\right)dz'dz\right].$$

From here, we see that, *if* the surpluses are the same ($S(z)=\mu(z)$), the two free-entry conditions are the same. That is, the investment margin is efficient. Because the investment margin is efficient, it might be tempting to conclude that the decentralized equilibrium is efficient.

However, the surpluses are *not* the same. That is, the valuation margin is inefficient. To see this fact, we can differentiate (15), impose $\gamma=\eta(\theta)$ and $\omega=1$, and evaluate at the steady state to obtain an ODE for the private surplus $S(z)$:

$$\left[r+\sigma+\eta(\theta)\zeta p(\theta)(1-G(z))\right]S'(z) = A,$$

with boundary condition $S(\underline{z})=0$, which we can solve as

$$S(z) = A\int_{\underline{z}}^{z}\frac{1}{r+\sigma+\eta(\theta)\zeta p(\theta)(1-G(\tilde{z}))}\,d\tilde{z}. \tag{24}$$

The reservation match quality is pinned down by the condition $S(\underline{z})=0$, which simplifies to

$$0 = A\underline{z} - h - A(1-\zeta)p(\theta)\int_{\underline{z}}^{\infty}\eta(\theta)\frac{1-G(\tilde{z})}{r+\sigma+\eta(\theta)\zeta p(\theta)(1-G(\tilde{z}))}\,d\tilde{z}. \tag{25}$$

Likewise, the ODE for the planner's surplus is

$$[r + \sigma + \zeta p(\theta)(1 - G(z))]\, \mu'(z) = A,$$

with boundary condition $\mu(\underline{z}^{SP}) = 0$, which we can solve as

$$\mu(z) = A \int_{\underline{z}^{SP}}^{z} \frac{1}{r + \sigma + \zeta p(\theta)(1 - G(\tilde{z}))}\, d\tilde{z}. \tag{26}$$

The planner's reservation match quality satisfies

$$0 = A\underline{z}^{SP} - h - A(1 - \zeta)p(\theta) \int_{\underline{z}^{SP}}^{\infty} \frac{\sigma\eta(\theta) + (1 - G(\tilde{z}))\zeta p(\theta)}{\sigma + (1 - G(\tilde{z}))\zeta p(\theta)} \frac{1 - G(\tilde{z})}{r + \sigma + \zeta p(\theta)(1 - G(\tilde{z}))}\, d\tilde{z}. \tag{27}$$

Comparing (24) and (26), for the same $(\theta, \underline{z})$, the private surplus is always larger than the planner's surplus due to the $\eta(\theta) < 1$ term in the denominator of the private surplus. Moreover, comparing (25) and (27), the planner's reservation match quality is always higher than the private reservation match quality, $\underline{z}^{SP} \geq \underline{z}$, because $\frac{\sigma\eta(\theta) + (1 - G(\tilde{z}))\zeta p(\theta)}{\sigma + (1 - G(\tilde{z}))\zeta p(\theta)} \geq \eta(\theta)$. The difference in the reservation match quality further widens the gap between the private and planner's surpluses. Therefore, for any given θ, the equilibrium overvalues jobs relative to the planner's valuation,

$$S(z) > \mu(z)$$

for any $z > \underline{z}$, as long as there is on-the-job search ($\zeta > 0$). The overvaluation of jobs, in turn, implies that jobs are excessively created and the market tightness is too high in the decentralized equilibrium:

$$\theta > \theta^{SP}.$$

The main intuition can be transparently seen by comparing (24) and (26). The denominator in each equation is the effective discount rate. By comparing the denominators, we can see that the private surplus discounts at a lower rate than the planner's surplus, due to the factor $\gamma = \eta(\theta) < 1$. The private surplus undervalues the loss from poaching: under Nash bargaining, only a fraction γ of that loss is internalized in the private surplus. By contrast, the planner fully takes into account the loss from poaching. This difference, in turn, results in the overvaluation of jobs and the excessive creation of vacancies in the decentralized equilibrium.

In what follows, we refine the mechanism and the intuition behind the inefficiency through a general version of our model.

3.3.2 Investment margin

We first consider the investment margin. In equilibrium, condition (17) represents the firm's incentive to invest in vacancies. In the planner's problem, the equation for this margin is (23). Let us compare these two equations. Suppose the Hosios condition (1) holds. If f_t^u and $f_t(z)$ are common and $\mu_t(z) = S_t(z)$, the equilibrium θ_t and the optimal θ_t coincide when the wage-setting protocol follows the sequential auction, that is, $\omega = 1$, as we have seen above. If the wage-setting rules are different from the sequential auction ($\omega < 1$), the efficiency of the equilibrium θ_t is not guaranteed even in this situation. For example, with Nash bargaining, that is, $\omega = 0$, the equilibrium value of θ_t is too high compared to the social optimum. Intuitively, this result is because of the *"worker-stealing" externality*: the poaching firm does not internalize the loss of the poached firm.[8] With the sequential auction, the poaching firm pays for the loss (not to the poached firm, but to the worker) in the form of higher wages.

In the existing literature, Gautier et al. (2010) emphasize a similar externality that arises in their model. However, Gautier et al. (2010) do not formally show the relationship

[8]This externality is analogous to the business-stealing externality in the economic growth literature.

between this externality and the sequential auction, as we do here. In addition, Gautier et al. (2010) do not consider the valuation margin, which we analyze in detail below. As we will see, the valuation margin is an essential element of our analysis, giving rise to inefficiency beyond the "worker-stealing externality." This analysis is entirely absent in the existing literature.

Note that from (18) and (19) and the definitions of f^u and $f_t(z)$ (i.e., (6)), the steady-state values of f^u and $f(z)$ are the same when θ is the same. Thus, now the important question for efficiency under the Hosios condition (and sequential auction in particular) is whether $\mu_t(z) = S_t(z)$ holds, which we will analyze next.

3.3.3 Valuation margin

To see how $\mu_t(z)$ and $S_t(z)$ are determined, let us look at the valuation margin. On the valuation margin, we rewrite the equation for the social planner (21) as:

$$\begin{aligned}(r+\sigma)\mu_t(z) - \dot{\mu}_t(z) = A_t z - h - p(\theta_t)\int_{\underline{z}_t}^{\infty} g(z')\mu_t(z')dz' \\ + \zeta p(\theta_t)\int_{z}^{\infty} g(z')(\mu_t(z') - \mu_t(z))dz' + \Delta^{SP}(\theta_t),\end{aligned} \tag{28}$$

where $\Delta^{SP}(\theta)$ is the term reflecting the net externality from forming an employment relationship:

$$\begin{aligned}\Delta^{SP}(\theta) \equiv & \underbrace{(1-\eta(\theta))\ p(\theta)\ \Omega_t}_{\text{positive externality of not searching off the job} \equiv \Delta^{SP,UE}(\theta)} \\ & \underbrace{-\zeta\ (1-\eta(\theta))\ p(\theta)\ \Omega_t}_{\text{negative externality of searching on the job} \equiv \Delta^{SP,EE}(\theta)},\end{aligned} \tag{29}$$

with

$$\Omega_t \equiv f_t^u \int_{\underline{z}_t}^{\infty} g(z')\mu_t(z')dz' + \int_{\underline{z}_t}^{\infty}\int_{\tilde{z}}^{\infty} f_t(\tilde{z})g(z')\left(\mu_t(z') - \mu_t(\tilde{z})\right)dz'd\tilde{z} \tag{30}$$

denoting the average gain from a meeting. The first term $A_t z - h$ on the right-hand side of (28) is the static gain from moving one unemployed worker to employment with the match quality z. The second term is the opportunity cost of not being able to conduct off-the-job search. This cost is offset by the possibility of on-the-job search, which is the last term in the first line.

The term $\Delta^{SP}(\theta)$ summarizes net externalities from forming an employment relationship. The first term inside $\Delta^{SP}(\theta)$ is the positive externality of not doing off-the-job search, which raises the probability of a match for all the other workers. The second term inside $\Delta^{SP}(\theta)$ is the negative externality of on-the-job search, which lowers the probability of a match for all the other workers as long as $\zeta > 0$. Note that when $\zeta = 1$ (employed workers have the same search efficiency as unemployed workers), the two externalities cancel out so that $\Delta^{SP}(\theta) = 0$. When $\zeta = 1$, workers impose the same congestion externality on the economy regardless of whether they are employed or unemployed. We will revisit this case later.

For the equilibrium, we rewrite (15) to facilitate comparison with (28):

$$\begin{aligned}(r+\sigma)S_t(z) - \dot{S}_t(z) = A_t z - h - p(\theta_t)\int_{\underline{z}_t}^{\infty} g(z')S_t(z')dz' \\ + \zeta p(\theta_t)\int_{z}^{\infty} g(z')\left(S_t(z') - S_t(z)\right)dz' + \Delta^{DE}(z,\theta_t),\end{aligned} \tag{31}$$

where

$$\begin{aligned}\Delta^{DE}(z,\theta) \equiv \quad & \underbrace{(1-\gamma)p(\theta)\int_{\underline{z}_t}^{\infty} g(z')S_t(z')dz'}_{\text{cost of not searching off the job that is unaccounted for} \equiv \Delta^{DE,UE}(z,\theta)} \\ & \underbrace{-\zeta(1-\gamma)p(\theta)\int_{z}^{\infty} g(z')\left(S_t(z') - \omega S_t(z)\right)dz'}_{\text{benefit of searching on-the-job that is unaccounted for} \equiv \Delta^{DE,EE}(z,\theta)}.\end{aligned} \tag{32}$$

The first line in (31) is analogous to that in the planner's solution (28). The term $\Delta^{DE}(z,\theta)$ summarizes the net opportunity cost that is unaccounted for by the match. The first term

inside $\Delta^{DE}(z,\theta)$ is the opportunity cost of off-the-job search that is unaccounted for. By being employed, a worker loses the opportunity for off-the-job search, but that opportunity cost shows up only as a γ fraction because a $(1-\gamma)$ fraction of the surplus accrues to the firm due to Nash bargaining. Similarly, the second term inside $\Delta^{DE}(z,\theta)$ is the benefit of on-the-job search that is unaccounted for. Among the total value of the new match $S_t(z')$, the amount $(1-\gamma)(S_t(z')-\omega S_t(z))$ accrues to *the new employer*, corresponding to (14). (The rest goes to the worker.) This amount is not accounted for when the "opportunity gain" of on-the-job search is counted in the surplus.

The comparison between $\Delta^{SP}(\theta)$ and $\Delta^{DE}(z,\theta)$ reveals these two are not necessarily equal even under the Hosios condition. In fact, this point is the fundamental reason that the efficiency in the valuation margin, $\mu_t(z) = S_t(z)$, is difficult to achieve. First and foremost, there is a fundamental difference between $\Delta^{SP}(\theta)$ and $\Delta^{DE}(z,\theta)$: $\Delta^{SP}(\theta)$ does not depend on z, while $\Delta^{DE}(z,\theta)$ does. A worker's on-the-job search imposes an externality on *all other* employed workers. The (unaccounted-for) opportunity cost in equilibrium is fundamentally about *the current match*. Consider a match with a high z. The private gain from on-the-job search on such a job is small, and thus, the unaccounted benefit is small. As a consequence, such a match has a large $\Delta^{DE}(z,\theta)$ compared to a low-z match. In contrast, the externality that one employed worker imposes on others is the same regardless of the value of z. Furthermore, we can observe that $\Delta^{DE}(z,\theta)$ is increasing in z, and thus, the gap between $\Delta^{DE}(z,\theta)$ and $\Delta^{SP}(\theta)$ is larger when z is larger. This relationship implies that the overvaluation of the match is particularly pronounced at the top of the job ladder.

The special case of $\zeta = 1$ provides a particularly clean comparison. When $\zeta = 1$, the net congestion externality is zero, $\Delta^{SP}(\theta) = 0$, because both employed and unemployed workers equally congest the matching market. By contrast, $\Delta^{DE}(z,\theta) \geq 0$. This inequality reflects that the imperfect appropriation of surplus from on-the-job search is smaller than that from off-the-job search because the surplus gain from on-the-job search is smaller

than that from off-the-job search.

3.4 The role of market segmentation

In the previous section, we highlighted the importance of two factors that impede efficiency along the valuation margin. First, moving a worker from unemployment to employment imposes externalities on both unemployed workers and other employed workers. Second, because employed workers are heterogeneous, offsetting the externalities to other workers by proportionally reducing the opportunity cost of employment for one worker is difficult.

To clarify these two factors further, in Appendix B, we analyze a model where (i) the matching functions for the unemployed and the employed are segmented, and (ii) for the employed, the matching functions for different values of z are segmented. We find that the equilibrium outcome is efficient with the Hosios condition and sequential auction ($\omega = 1$).[9] Both are needed for efficiency—just segmenting the unemployed market and the employed market is not sufficient. The results are analogous to the efficiency result in Menzio and Shi (2011), who show the efficiency of a job-ladder model with directed search. Our model features random search, and for efficiency, in addition to segmentation, the Hosios condition and sequential auction mechanism are required.

4 Analytical characterizations

We now demonstrate several analytical characterizations. First, we show that, under the Hosios condition, equilibrium vacancy creation is always excessive. Second, we show that the efficient allocation can be implemented by a combination of employment and vacancy taxes and characterize the properties of the optimal taxes.

[9] We thank Xincheng Qiu for suggesting this exercise.

4.1 Excessive vacancy creation

In Appendix C, we provide a useful intermediate result: the steady-state decentralized equilibrium and planner's allocation can each be reduced to two equations in two unknowns $(\theta, \underline{z})$. Building on these characterizations, we can derive a clear-cut prediction on the direction of the inefficiency under the Hosios condition.

Proposition 1 *Suppose $\gamma = \eta(\theta^{SP})$. Then, the steady-state equilibrium market tightness θ in the decentralized equilibrium is higher than the socially efficient level when $\zeta > 0$.*

Proofs of this and the remaining propositions are collected in Appendix E. There, we prove a more general version of the result: there exists $\hat{\gamma} > \eta(\theta^{SP})$ such that if $\gamma < \hat{\gamma}$, the steady-state equilibrium market tightness θ in the decentralized equilibrium is higher than the socially efficient level.

Section 3.3.1 showed that, under sequential auction ($\omega = 1$), the investment margin is efficient, but the inefficiency along the valuation margin leads to overvaluation of jobs and excessive vacancy creation. Proposition 1 generalizes the excessive-vacancy-creation result beyond sequential auction. As discussed earlier, with imperfect offer-matching, $\omega < 1$, the investment margin also becomes inefficient because private agents fail to internalize the business-stealing externality from vacancy creation. This other force further exacerbates the excessive vacancy creation.

Proposition 1 is important, particularly in the context of policy evaluations. It implies that assuming the Hosios condition is not "neutral" in a job-ladder model, as it automatically favors a policy that discourages vacancy posting. This result highlights the fundamental difference from the model without on-the-job search, where the Hosios condition implies no intervention is optimal. Beyond isolating the role of on-the-job search as a source of inefficiency, Proposition 1 is directly relevant for existing applied work, as it has been common to impose the Hosios condition in the calibration of the quantitative DMP

literature.[10] In the case of a job-ladder model, this calibration implies that policies that discourage vacancy posting improve efficiency even in the absence of other distortions.

4.2 Decentralization

We now turn to the implementation of the efficient allocation. We show that the employment tax, which corrects for the valuation margin, and the vacancy tax, which corrects for the investment margin, implement the efficient allocation.[11]

The government imposes taxes on workers and firms. The government imposes a vacancy tax χ^e so that the vacancy-posting cost that the firms face is given by $\kappa + \chi_t^e$. This tax replaces κ with $\kappa + \chi_t^e$ in (17). The government also imposes a tax of $\tau(z)$ on employed workers. As a result, the flow value $A_t z - h$ is replaced by $A_t z - h - \tau(z)$ in the joint match surplus (15). The government finances these fiscal instruments through a lump-sum tax on workers.

In order for the decentralized equilibrium to achieve the efficient allocation, the tax on employed workers has to satisfy

$$\tau(z) = \Delta^{DE}(z, \theta_t^{SP})|_{S=\mu} - \Delta^{SP}(\theta_t^{SP}), \tag{33}$$

where $\Delta^{DE}(z, \theta_t^{SP})|_{S=\mu}$ evaluates (32) at θ_t^{SP} and $S = \mu$ and $\Delta^{SP}(\theta_t^{SP})$ evaluates (29) at θ_t^{SP}. This employment tax corrects for the valuation margin. The vacancy tax has to satisfy

$$\chi_t^e = (\eta(\theta_t^{SP}) - \gamma) q(\theta_t^{SP}) \Omega_t + (1-\omega)(1-\gamma) q(\theta_t^{SP}) \int_{\underline{z}_t}^{\infty} \int_{z}^{\infty} f_t(z) g(z') \mu_t(z) dz' dz, \tag{34}$$

where Ω_t is defined in (30). The vacancy tax here corrects for the investment margin. The following result summarizes the implementation.

[10] In models without on-the-job search, this calibration has been frequently used since Shimer (2005). Recent examples that employ this calibration strategy in a model with on-the-job search are Engbom (2019) and Faberman et al. (2022).

[11] Appendix D studies an alternative implementation with unemployment insurance and a vacancy tax.

Proposition 2 *Suppose that the taxes* $\{\tau(z), \chi_t^e\}$ *are set so that* (33) *and* (34) *hold. Then, the decentralized equilibrium implements the efficient allocation. In such an equilibrium,* $S_t(z) = \mu_t(z)$ *for all* z.

The proposition demonstrates that the decentralized equilibrium with taxes implements an efficient allocation and that the match surplus is valued in the same manner as the social planner's solution.

The following proposition characterizes the properties of these wedges under the Hosios condition.

Proposition 3 *Suppose that the Hosios condition* $\gamma = \eta(\theta^{SP})$ *holds and there is on-the-job search, that is,* $\zeta > 0$. *Then, the following holds:*

1. *The employment tax* $\tau(z)$ *is increasing in* z *and positive,* $\tau(z) \geq 0$.

2. *The vacancy tax is positive,* $\chi^e \geq 0$. *It is zero when offer matching is perfect,* $\omega = 1$.

Recall that the employment tax corrects the valuation margin: it equals the gap $\Delta^{DE} - \Delta^{SP}$ between the unaccounted private opportunity cost and the planner's net externality. Under the Hosios condition, the employment tax is positive, reflecting the fact that jobs are overvalued everywhere on the job ladder. The employment tax rises with job-ladder z, reflecting the fact that jobs are particularly overvalued at the top of the job ladder. Intuitively, the unaccounted benefit of on-the-job search declines with job-ladder rank, while the congestion externality they impose remains the same. Therefore, imperfect appropriation through bargaining fails to correct for the congestion externality especially at the top of the job ladder.

Also recall that the vacancy tax corrects the investment margin. Under the Hosios condition, it reduces to the worker-stealing term and is therefore zero when offer matching is perfect ($\omega = 1$). When $\omega < 1$, the vacancy tax is positive because the worker-stealing externality induces over-creation of vacancies and requires a positive tax to correct for it.

Parameter	Description	Value	Target/Source
PANEL A. ASSIGNED PARAMETERS			
r	Discount rate	0.004	Annual interest rate 5%
η	Elasticity of matching function	0.5	Standard
γ	Worker bargaining power	0.5	Standard
ω	Offer matching	1.0	Dey and Flinn (2005), Cahuc et al. (2006)
h	Home production value	0.6	Normalization
A	Aggregate productivity	1.0	Normalization
PANEL B. CALIBRATED PARAMETERS			
σ	Separation rate	0.024	EU rate 2.4%
α	Pareto distribution shape parameter	7.49	$h/\mathbb{E}[w] = 0.4$
κ	Cost of vacancy creation	0.40	Market tightness 1
$m\, z_{\min}^{\alpha}$	Composite matching efficiency	1.84	Unemployment rate 5.5%
ζ	On-the-job search efficiency	0.24	EE rate 2.5%

Table 1: Parameter values

5 Quantitative exploration

In this section, we quantitatively explore the difference between the efficient allocation and the equilibrium outcome.

5.1 Calibration

Table 1 summarizes our parameterization. We interpret one period as a month and set the discount rate to 5% annually, $r = 0.05/12$. The exogenous separation rate is set to 2.4% at a monthly frequency, that is, $\sigma = 0.024$. We assume the Cobb-Douglas matching function, $M(x, v) = mx^{\eta}v^{1-\eta}$ with $\eta = 0.5$, and the bargaining power is such that $\gamma = \eta$ (i.e., the Hosios condition). This assumption ensures that without on-the-job search, the equilibrium is efficient, thereby isolating the role of on-the-job search as a source of inefficiency. We normalize units of vacancies so that $\theta \equiv v/[u + \zeta(1-u)] = 1$ in the steady state. We assume the productivity distribution is given by the Pareto distribution: $G(z) = 1 - (z/z_{min})^{-\alpha}$, which is consistent with the notion of a balanced growth path in Martellini and Menzio (2020). The flow value of unemployment is normalized to $h = 0.6$,

and the aggregate productivity is normalized to $A = 1$.[12] We set the offer-matching parameter to $\omega = 1$, which corresponds to the sequential-auction protocol of Dey and Flinn (2005) and Cahuc et al. (2006) and is widely used as a benchmark in the subsequent literature. The matching efficiency m and Pareto location parameter $z_{\min}$ always enter through $mz_{\min}^{\alpha}$, so we treat them as a single composite parameter.

We then choose four parameters $\{\alpha, mz_{min}^{\alpha}, \kappa, \zeta\}$ to jointly match the following three empirical moments, together with the tightness normalization: the steady-state unemployment rate of 5.5%; the monthly employment-to-employment transition rate of 2.5%; and the ratio of the flow value of unemployment to the average wage, $h/\mathbb{E}[w] = 0.40$, in the spirit of the replacement-rate target in Shimer (2005), where $\mathbb{E}[w]$ is the mean wage among the employed.

Although all parameters are jointly calibrated, we provide a heuristic argument for how each moment identifies each parameter. First, the composite matching-efficiency parameter $mz_{\min}^{\alpha}$ is identified from the steady-state unemployment rate. The efficiency of on-the-job search, ζ, is inferred from the employment-to-employment transition rate. The cost of vacancy creation, κ, is chosen to ensure our normalization of $\theta = 1$. Since the Pareto tail parameter α controls the thickness of the match quality above the reservation match quality, it most naturally affects the ratio of the flow value of unemployment to average wages $h/\mathbb{E}[w]$.

5.2 Decentralized equilibrium vs. efficient allocation

Now, we compare the outcomes of the decentralized equilibrium and the efficient allocation. Table 2 compares the aggregate variables. The equilibrium allocation has too many vacancies; θ is too high, which is due to the overvaluation of jobs. The overvaluation of jobs also makes the equilibrium less selective in forming a match, resulting in a lower

[12] With a Pareto match-quality distribution and perfect offer matching ($\omega = 1$), z_{min}, κ, and h are not separately identified, because any combination of (z_{min}, h, κ) with the same z_{min}/h and κ/h results in the same labor market allocation. Therefore, the choice of h merely amounts to normalization.

	Equilibrium	Efficient
Market tightness	1.00	0.78
Reservation match quality	1.221	1.228
Consumption	1.4624	1.4633
UE rate	0.412	0.349
EE rate	0.025	0.023
Unemployment rate	5.5%	6.4%
Average match quality	1.634	1.620

Table 2: Comparison of equilibrium and efficient allocations

Note: The table shows aggregate outcomes in the decentralized equilibrium and the efficient allocation. Average match quality is computed among the employed.

reservation match quality in the decentralized equilibrium than in the planner's solution. The excessive vacancy creation and lower reservation match quality both imply the unemployment rate is too low in the decentralized equilibrium. The unemployment rate is 0.9 percentage points higher in the efficient allocation. The difference in the unemployment rate, therefore, is economically significant. The resulting consumption is lower in the equilibrium allocation, although the difference (about 0.06%) is small.

The last row of Table 2 shows that the average match quality in the decentralized equilibrium is higher than in the planner's solution.[13] As the market tightness θ is higher in the decentralized equilibrium, workers tend to climb the job ladder faster and consequently tend to concentrate at the top of the job ladder. Interestingly, the average match quality is higher than in the planner's solution despite the fact that the planner is more selective in forming a match, reflected in the higher reservation match quality. The result of too many productive jobs in the decentralized equilibrium contrasts with models featuring ex-ante job heterogeneity, such as Acemoglu (2001), where too few "good jobs" tend to exist in equilibrium compared to the efficient allocation.

5.3 Steady-state comparative statics

This section conducts two comparative statics in terms of key parameters.

[13] Appendix Figure K.4 plots the employment distribution in the equilibrium and in the efficient allocation.

Figure 1a varies the degree of on-the-job search ζ. The other parameters are kept constant at their baseline values. Because we impose the Hosios condition, the decentralized equilibrium is efficient when $\zeta = 0$ (no on-the-job search). What is striking here is that the labor market tightness in the decentralized equilibrium significantly departs from the optimal outcome even with a small amount of on-the-job search; for example, $\zeta = 0.1$.

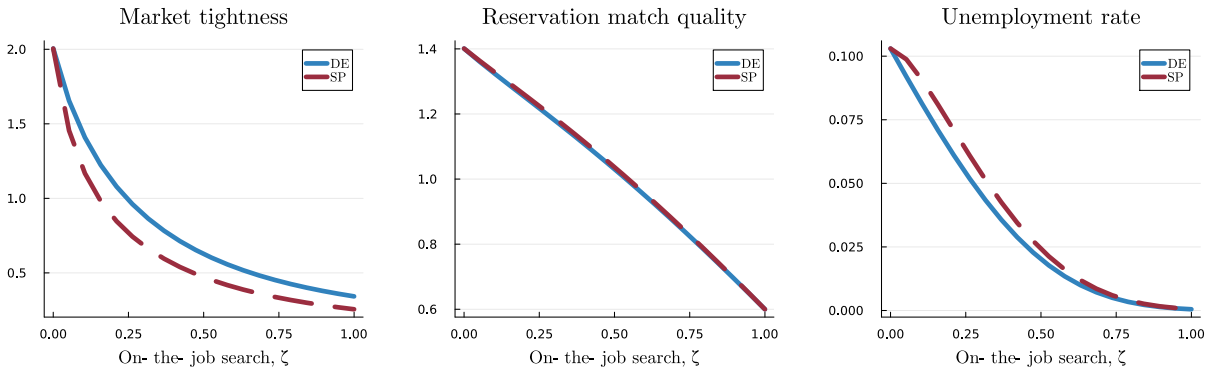
Figure 1b varies the degree of offer matching ω. We have seen that our benchmark, $\omega = 1$, achieves optimal investment in vacancies when valuation is correct. Therefore, it is not surprising that the equilibrium allocation tends to be farther from the efficient outcome when ω is lower. The equilibrium labor market tightness decreases with ω, and the equilibrium reservation match quality $\underline{z}$ increases slightly. Both effects raise the equilibrium unemployment rate. The difference in the equilibrium unemployment rate between $\omega = 0$ and $\omega = 1$ is approximately 0.9 percentage points, indicating that the worker-stealing externality is economically significant. Note that the efficient allocation is invariant to ω.

Why does the equilibrium market tightness fall with ω? On the one hand, a higher ω means a poaching firm must pay a higher wage, which discourages vacancy posting for a given $S(z)$. On the other hand, the valuation-margin equation (15) reveals that a higher ω raises $S(z)$. This result follows because workers are forward-looking: workers' surplus (and thus the current match surplus) includes the future possibility of offer matching and counter-offers. In our calibration, the former effect dominates, and the market tightness falls with ω.

5.4 Transition dynamics

So far, our focus has been on the steady state. We now consider the economy's response to a productivity shock. This experiment is important in evaluating how the efficiency properties interact with various phases of the business cycle.

We consider a one-time unanticipated shock ("MIT shock") to the aggregate produc-



(a) Varying the on-the-job search parameter ζ



(b) Varying the offer-matching parameter ω

Figure 1: Steady-state comparative statics: equilibrium vs. planner

tivity, A_t, that increases by 1% at $t = 0$, which decays with rate ρ_A:

$$d \ln A_t = -\rho_A \ln(A_t / A) dt.$$

We set $\rho_A = 0.04$, which corresponds to the autocorrelation of 0.96 at the monthly frequency. We compare the response of the social planner's solution with that of the decentralized equilibrium, starting from the steady state described earlier. Appendix F shows that, given the productivity path and the initial employment distribution, solving for the transition dynamics amounts to solving for the sequence of market tightness $\{\theta_t\}_{t \geq 0}$, without the need to solve directly for the employment distribution. For each candidate tightness path, we solve the HJB backward to obtain surplus and the endogenous cutoff,

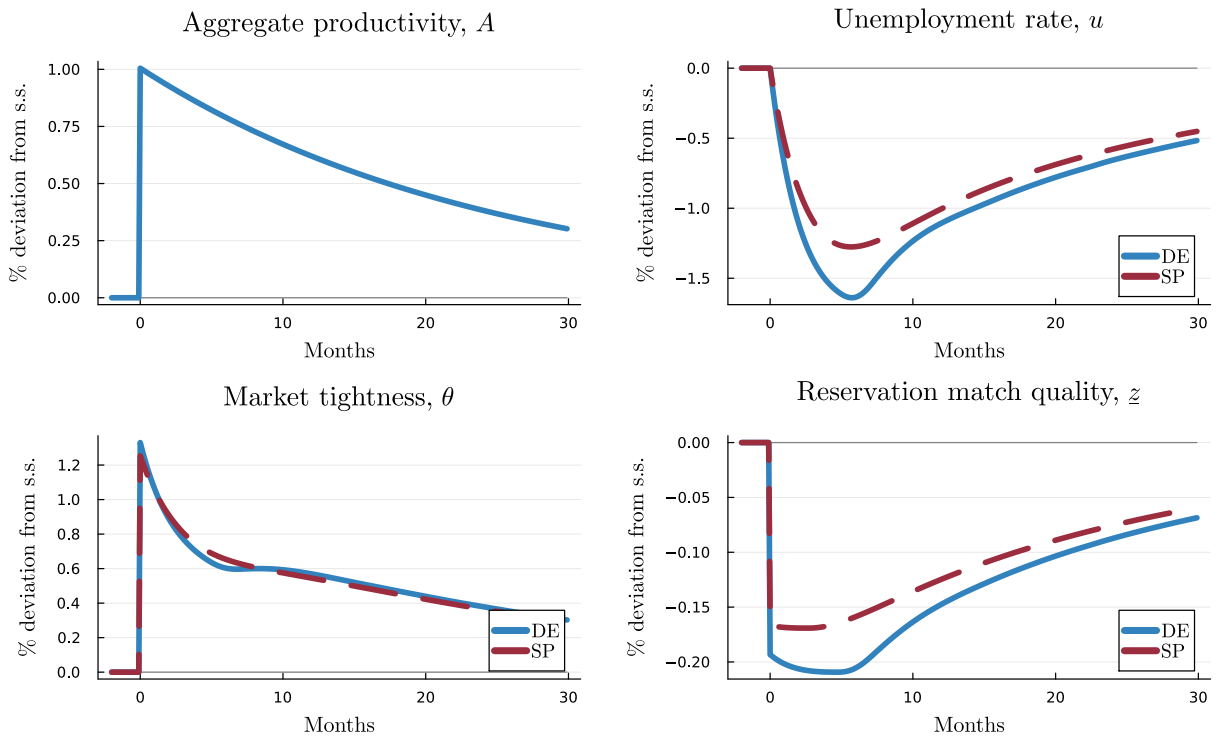


Figure 2: Response to productivity shock: Equilibrium and planner solution

Note: The figure plots the response of the decentralized equilibrium and the social planner's solution to a 1% initial increase in aggregate productivity at $t = 0$. All panels are percent deviations from each allocation's own steady state. The solid line represents the decentralized equilibrium, and the dashed line represents the social planner's solution.

then solve for the employment distribution forward. We update the tightness path to satisfy free entry, solving the resulting system nonlinearly.

The top-left panel of Figure 2 shows the path of aggregate productivity. The top-right panel shows that, over the first half-year after the shock, the unemployment rate decreases more in the decentralized equilibrium than in the planner's solution. Given that the steady-state unemployment rate was already lower in the equilibrium than in the planner's solution, the positive productivity shock exacerbates the gap in the unemployment rate. The bottom two panels show that this difference is driven by a slightly larger initial rise in market tightness and a persistently larger fall in reservation match quality.

What underlies these two patterns is that the overvaluation of jobs is further exacerbated in response to positive productivity shocks. This result holds because the ineffi-

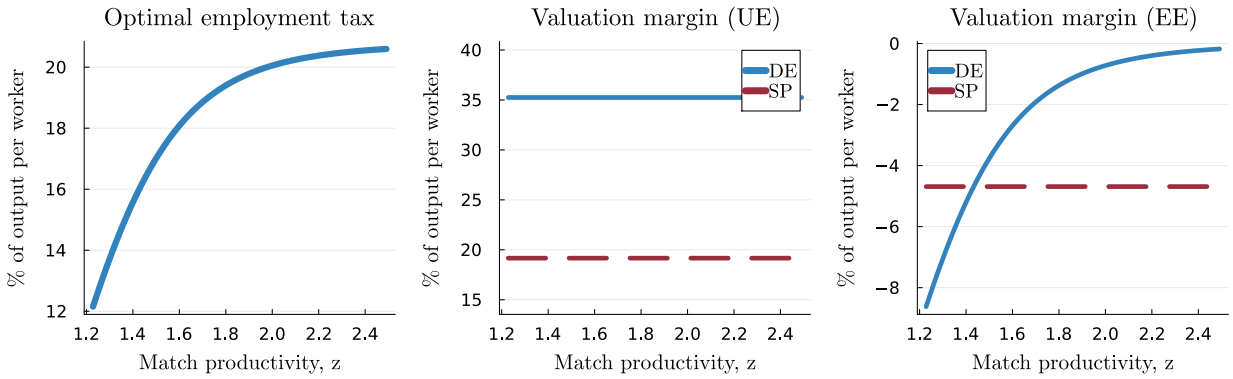


Figure 3: Optimal employment tax and valuation-wedge decomposition

Note: The left panel plots the optimal employment tax $\tau(z) = \Delta^{DE}(z, \theta^{SP})|_{S=\mu} - \Delta^{SP}(\theta^{SP})$ in (33). The middle panel plots $\Delta^{DE,UE}$ in (32) evaluated at $S = \mu$ and $\Delta^{SP,UE}$ in (29). The right panel plots $\Delta^{DE,EE}(z)$ and $\Delta^{SP,EE}$. All objects are evaluated at the planner's steady-state allocation and expressed as a fraction of average output per worker.

ciency along the valuation margin scales with the job-finding rate, $p(\theta)$. Consequently, the rise in tightness after the shock amplifies overvaluation, which lowers reservation match quality and raises the market tightness along the transition.

5.5 Decentralization

We now quantitatively illustrate the decentralization of the efficient allocation via the wedge-correcting instruments characterized in Section 4.2. (The quantitative magnitudes of the simple UI and vacancy-tax instruments are reported in Appendix D.)

Figure 3 plots the employment tax and the associated valuation-wedge decomposition.[14] The left panel plots the employment tax (33) that closes the overall valuation wedge. In line with Proposition 3, $\tau(z)$ is positive and increasing in z. Quantitatively, the overvaluation of jobs is substantial toward the top of the job ladder, reaching 20% of the average output per worker.

The middle and right panels of Figure 3 decompose the overall wedge into UE and EE components of the valuation margin Δ^{DE} and Δ^{SP}, discussed in Section 3.3.3. The middle

[14] Appendix Figure K.5 plots the level of the match surplus in the decentralized equilibrium and the planner's solution evaluated at the planner's steady-state tightness θ^{SP}.

panel shows the value of off-the-job search unaccounted for by the private agent (DE) and the positive externality from not doing off-the-job search (SP). The presence of on-the-job search reduces the gains from meeting and thereby the congestion externality, whereas the imperfect appropriation of surplus from off-the-job search remains substantial. This comparison implies that the social value of not doing off-the-job search is smaller than the private value.

The right panel shows the value of on-the-job search unaccounted for by the private agent (DE) and the negative externality from on-the-job search (SP). In equilibrium, a worker with high z has small job-to-job gains, and thus, the "missing private gain" due to the Nash bargaining is not large (in absolute value), and the opposite is true for a worker with low z. By contrast, because of the random-search assumption, the externality one person imposes due to on-the-job search is the same regardless of the value of z. For workers with large z, therefore, the effect of the negative externality they impose on others dominates the imperfect appropriation of surplus from on-the-job search.

Finally, note that the optimal vacancy tax is zero, as we assume the Hosios condition and sequential auction (see Proposition 3).

5.6 Robustness

In Appendix I, we explore the robustness of our results under several alternative calibrations and parameterizations. First, we consider an alternative calibration with a higher replacement rate $h/\mathbb{E}[w] = 0.6$. Second, we parameterize our model so that all matches on the support are accepted, in which case the reservation match quality margin is absent. Third, we assume the match quality is log-normally distributed rather than Pareto distributed. In all cases, the overall patterns are very similar to the baseline calibration. The decentralized equilibrium continues to feature an unemployment rate that is 0.8–1.0 percentage points below the efficient level.

6 Conclusion

This paper examines the efficiency of the decentralized equilibrium in a DMP-style job-ladder model. We identify two margins where inefficiencies can arise: the investment margin and the valuation margin. In the presence of on-the-job search, the Hosios condition does not ensure the efficiency of the decentralized equilibrium. Even when we impose a wage-setting protocol that ensures the investment margin remains undistorted, the valuation margin remains distorted. We show that, for a wide range of wage-setting protocols, too many vacancies are posted in the decentralized equilibrium under the Hosios condition. These results have important implications for policy evaluations that utilize job-ladder models.

The quantitative exercise focuses on a situation where the economy is calibrated as in the standard macroeconomic literature, and the Hosios condition is satisfied. It reveals that even with a small amount of on-the-job search, the optimal unemployment rate and the equilibrium unemployment rate can differ significantly. We also examine how efficient allocations can be decentralized.

The intuitions about inefficiencies would carry over to more complex models with various other factors. An important factor is how labor markets for different workers are segmented. The segmentation can be in different dimensions—here, we focused on the employment status (unemployed vs. employed) and the match quality. In practice, the labor market can be segmented across various other dimensions, such as gender, race, education, experience, and geography. Our paper shows that understanding the degree of segmentation in the economy is essential to designing policies under a frictional labor market.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Cursor, ChatGPT, and Claude to check grammatical and typographical errors, to receive editorial suggestions, to check proofs and algebra, and to assist with the simulation analysis. After using these tools/services, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Appendix

A Pissarides (1985) model

In this appendix, we outline the entire Pissarides (1985) model and derive the Hosios condition by comparing the social planner's solution with the market outcome.

A.1 Social planner's problem

The social planner's problem is

$$\max_{n_t,\theta_t} \int_0^\infty e^{-rt}[zn_t + h(1-n_t) - \kappa\theta_t(1-n_t)]dt$$

subject to

$$\dot{n}_t = p(\theta_t)\,(1-n_t) - \sigma n_t.$$

The current-value Hamiltonian for this problem can be written as

$$H = zn_t + h\,(1-n_t) - \kappa\theta_t\,(1-n_t) + \mu_t\,(p(\theta_t)\,(1-n_t) - \sigma n_t)\,.$$

Thus, first-order conditions are

$$\kappa(1-n_t) = p'(\theta_t)\mu_t(1-n_t) \tag{35}$$

and

$$z - h + \kappa\theta_t - p(\theta_t)\mu_t - (r+\sigma)\mu_t + \dot{\mu}_t = 0. \tag{36}$$

From (35), noting $p'(\theta) = (1 - \eta(\theta))q(\theta)$,

$$\kappa = (1 - \eta(\theta))q(\theta)\mu_t.$$

This equation corresponds to (2) in the main text. From (36) in the steady state where $\dot{\mu}_t = 0$,

$$(r + \sigma)\mu_t = z - h - (p(\theta_t)\mu_t - \kappa\theta_t)$$

holds. This equation corresponds to (4) in the main text.

A.2 Decentralized equilibrium

In the decentralized equilibrium, the Hamilton-Jacobi-Bellman (HJB) equation for an unemployed worker is

$$rU_t = h + p(\theta_t)(W_t - U_t) + \dot{U}_t, \tag{37}$$

where U_t is the value of an unemployed worker and W_t is the value of an employed worker. The HJB equation for an employed worker is

$$rW_t = w_t - \sigma(W_t - U_t) + \dot{W}_t, \tag{38}$$

where w_t is the wage at time t. On the firm side, the HJB equation for a matched job is

$$rJ_t = z - w_t - \sigma(J_t - V_t) + \dot{J}_t, \tag{39}$$

where J_t is the value of a matched job and V_t is the value of a vacancy. The value of a vacancy satisfies

$$rV_t = -\kappa + q_t(J_t - V_t) + \dot{V}_t. \tag{40}$$

We assume anyone can post a vacancy (free entry). Therefore, the value of a vacancy is driven down to zero in an equilibrium where the vacancy posting is strictly positive (we focus on such an equilibrium):

$$V_t = 0. \tag{41}$$

The wages are determined by Nash bargaining, which implies

$$W_t = \gamma S_t + U_t \tag{42}$$

and

$$J_t = (1-\gamma) S_t + V_t, \tag{43}$$

where $S_t \equiv W_t + J_t - U_t - V_t$ is the joint surplus from a match, and γ is workers' bargaining power. Thus, an employed worker and a matched firm share the surplus with the fractions γ and $(1-\gamma)$, in addition to their outside options. Using (41) and (43) in (40), we obtain

$$\kappa = (1-\gamma) q(\theta_t) S_t.$$

This equation corresponds to (3) in the main text. Adding (38) and (39) and subtracting (37) and using (41) and (42) (also imposing the steady-state condition $\dot{S}_t = 0$), we obtain

$$(r+\sigma) S_t = z - h - p(\theta_t) \gamma S_t,$$

which corresponds to (5) in the main text.

B Segmented market

We modify our baseline model by assuming the labor market is segmented by workers' employment status as well as the match quality of current jobs if workers are employed. We denote the market tightness that unemployed workers face as θ^u and the market tightness that employed workers with match quality z face as $\theta^e(z)$. There is free entry in each of the segmented markets. The outside option is revalued between meetings as in the baseline model, and $\dot{W}_t(z,\bar{O})$ and $\dot{J}_t(z,\bar{O})$ include this revaluation.

The value of being unemployed is given by

$$rU_t = h + p(\theta_t^u)\int g(z)\max\{W_t(z,0) - U_t, 0\}dz + \dot{U}_t.$$

The value function of workers employed with match quality z and an outside option $\bar{O} \in [0, \omega S_t(z)]$ is given by

$$\begin{aligned} rW_t(z,\bar{O}) = {} & w_t(z,\bar{O}) \\ & + \zeta p(\theta_t^e(z))\int_z^\infty g(z')\left(W_t(z',\omega S_t(z)) - W_t(z,\bar{O})\right)dz' \\ & + \zeta p(\theta_t^e(z))\int_0^z g(z')\left(W_t(z,\max\{\omega S_t(z'),\bar{O}\}) - W_t(z,\bar{O})\right)dz' \\ & + \sigma(U_t - W_t(z,\bar{O})) + \dot{W}_t(z,\bar{O}). \end{aligned}$$

The value of a filled job with productivity z and outside option $\bar{O} \in [0, \omega S_t(z)]$ is given by

$$\begin{aligned} rJ_t(z,\bar{O}) = {} & A_t z - w_t(z,\bar{O}) \\ & + \zeta p(\theta_t^e(z))\int_z^\infty g(z')\left(V_t - J_t(z,\bar{O})\right)dz' \\ & + \zeta p(\theta_t^e(z))\int_0^z g(z')\left(J_t(z,\max\{\omega S_t(z'),\bar{O}\}) - J_t(z,\bar{O})\right)dz' \\ & + \sigma\left(V_t - J_t(z,\bar{O})\right) + \dot{J}_t(z,\bar{O}). \end{aligned}$$

The value of a vacancy in the market for unemployed workers is given by

$$rV_t^u = -\kappa + q(\theta_t^u) \int_0^\infty g(z) \max\{J_t(z,0) - V_t, 0\} dz + \dot{V}_t^u,$$

and the value of a vacancy in the market for employed workers with match quality z is given by

$$rV_t^e(z) = -\kappa + q(\theta_t^e(z)) \int_z^\infty g(z') \max\{J_t(z', \omega S_t(z)) - V_t, 0\} dz' + \dot{V}_t^e(z).$$

The free-entry condition is

$$V_t = \max\{V_t^u, \max_z V_t^e(z)\} = 0.$$

With Nash bargaining, the surplus from a match is

$$\begin{aligned}(r+\sigma)S_t(z) = A_t z - h \\ + \zeta p(\theta_t^e(z)) \int_z^\infty g(z') \left(\omega S_t(z) + \gamma \left[S_t(z') - \omega S_t(z)\right] - S_t(z)\right) dz' \\ - p(\theta_t^u) \int_{\underline{z}_t}^\infty g(z') \gamma S_t(z') dz' + \dot{S}_t(z)\end{aligned} \tag{44}$$

for $z \geq \underline{z}_t$, where $\underline{z}_t$ satisfies

$$S_t(\underline{z}_t) = 0.$$

The matches with $z < \underline{z}_t$ are not formed.

The free-entry conditions are (assuming all markets have strictly positive vacancy postings)

$$\kappa = (1-\gamma) q(\theta_t^u) \int_{\underline{z}_t}^\infty g(z) S_t(z) dz \tag{45}$$

and

$$\kappa = (1-\gamma)q(\theta_t^e(z))\int_z^\infty g(z')\left[S_t(z') - \omega S_t(z)\right]dz'. \tag{46}$$

Now, we turn to the social planner's problem. The social planner's problem is to choose $\{\theta_t^u, \theta_t^e(z), n_t(z), \mathbb{I}_t^{UE}(z), \mathbb{I}_t^{EE}(z,z'), \varsigma_t(z)\}$ to maximize

$$\int_0^\infty e^{-rt}\left[\int_0^\infty A_t z n_t(z)dz + h\left(1-\int_0^\infty n_t(z)dz\right) - \kappa\theta_t^u\left(1-\int_0^\infty n_t(z)dz\right) - \kappa\int_0^\infty \theta_t^e(z)\zeta n_t(z)dz\right]dt$$

subject to

$$\begin{aligned}\dot{n}_t(z) = &\left(1-\int n_t(z')dz'\right)p(\theta_t^u)g(z)\mathbb{I}_t^{UE}(z) + \int_0^\infty p(\theta_t^e(z'))g(z)\mathbb{I}_t^{EE}(z',z)\zeta n_t(z')dz' \\ &- \int_0^\infty p(\theta_t^e(z))g(z')\mathbb{I}_t^{EE}(z,z')\zeta n_t(z)dz' - \sigma n_t(z) - \varsigma_t(z).\end{aligned}$$

The current-value Hamiltonian for this problem is

$$\begin{aligned}H = &\int_0^\infty A_t z n_t(z)dz + h\left(1-\int_0^\infty n_t(z)dz\right) - \kappa\theta_t^u\left(1-\int_0^\infty n_t(z)dz\right) - \kappa\int_0^\infty \theta_t^e(z)\zeta n_t(z)dz \\ &+ \int_0^\infty \mu_t(z)\left[\left(1-\int_0^\infty n_t(z')dz'\right)p(\theta_t^u)g(z)\mathbb{I}_t^{UE}(z) + \zeta\int_0^\infty p(\theta_t^e(z'))g(z)\mathbb{I}_t^{EE}(z',z)n_t(z')dz'\right. \\ &\left. - \zeta\int_0^\infty p(\theta_t^e(z))g(z')n_t(z)\mathbb{I}_t^{EE}(z,z')dz' - \sigma n_t(z) - \varsigma_t(z)\right]dz,\end{aligned}$$

where $\mu_t(z)$ is the costate variable that represents the shadow value of the distribution constraint above. Thus, $\mu_t(z)$ is the shadow value of creating one additional match with match quality z.

The optimality condition for $\{\mathbb{I}_t^{UE}(z), \mathbb{I}_t^{EE}(z,z')\}$ is

$$\mathbb{I}_t^{UE}(z) = \begin{cases} 1 & \mu_t(z) > 0 \\ 0 & \mu_t(z) \leq 0 \end{cases},$$

and

$$\mathbb{I}_t^{EE}(z,z') = \begin{cases} 1 & \mu_t(z') > \mu_t(z) \\ 0 & \mu_t(z') \leq \mu_t(z) \end{cases}.$$

The pointwise optimality condition for the nonnegative endogenous separation flow $\varsigma_t(z)$ implies $\mu_t(z) \geq 0$ for all z with $n_t(z) > 0$.

The first-order optimality condition on $n_t(z)$ is

$$\begin{aligned}(r+\sigma)\mu_t(z) = A_t z - h - \int_{\underline{z}_t}^{\infty} p(\theta_t^u) g(z') \mu_t(z') dz' + \zeta p(\theta_t^e(z)) \int_z^{\infty} g(z')(\mu_t(z') - \mu_t(z)) dz' \\ + \kappa\theta_t^u - \kappa\zeta\theta_t^e(z) + \dot{\mu}_t(z),\end{aligned} \tag{47}$$

where we have already imposed the fact that $\mu_t(z)$ is increasing in z. The reservation match quality $\underline{z}_t$ satisfies

$$\mu_t(\underline{z}_t) = 0.$$

The first-order optimality condition for θ_t^u is

$$\kappa = (1 - \eta(\theta_t^u)) q(\theta_t^u) \int_{\underline{z}_t}^{\infty} \mu_t(z') g(z') dz' \tag{48}$$

The first-order condition with respect to $\theta_t^e(z)$ is

$$\begin{aligned}\kappa = (1 - \eta(\theta_t^e(z))) q(\theta_t^e(z)) \int_0^{\infty} \mu_t(z') g(z') \mathbb{I}_t^{EE}(z,z') dz' \\ - (1 - \eta(\theta_t^e(z))) q(\theta_t^e(z)) \int_0^{\infty} g(z') \mathbb{I}_t^{EE}(z,z') dz' \mu_t(z),\end{aligned} \tag{49}$$

which we can rewrite as

$$\kappa = (1 - \eta(\theta_t^e(z))) q(\theta_t^e(z)) \int_z^{\infty} g(z') \left[\mu_t(z') - \mu_t(z)\right] dz'.$$

Using free-entry conditions, we can rewrite (47) as

$$(r+\sigma)\mu_t(z) = A_t z - h - \eta(\theta_t^u)p(\theta_t^u)\int_{\underline{z}_t}^{\infty} g(z')\mu_t(z')dz' + \zeta\eta(\theta_t^e(z))p(\theta_t^e(z))\int_z^{\infty} g(z')(\mu_t(z') - \mu_t(z))dz' + \dot{\mu}_t(z). \quad (50)$$

Comparing (44), (45), and (46) with (50), (48), and (49), we can see that the social planner's problem is equivalent to the decentralized equilibrium as long as the Hosios condition holds, $\eta(\theta) = \gamma$ for all θ, and offer matching is perfect, $\omega = 1$. We summarize the results as follows.

Proposition 4 *Consider the environment with segmented labor markets described above. The decentralized equilibrium is efficient if the Hosios condition holds, that is, $\eta(\theta) = \gamma$ for all θ, and offer matching is perfect, $\omega = 1$.*

C Steady-state characterizations

We show that the steady-state decentralized equilibrium can be simply characterized with two equations with two unknowns. Toward this end, we evaluate (15) in the steady state and take the derivative with respect to z to obtain the following ODE,

$$[r+\sigma+(1-\omega(1-\gamma))\zeta p(\theta)(1-G(z))]\,S'(z) \;=\; A+(1-\gamma)(1-\omega)\zeta p(\theta)g(z)S(z),$$

with a boundary condition $S(\underline{z}) = 0$. The reservation match quality $\underline{z}$ satisfies

$$0 = A\underline{z} - h - \gamma(1-\zeta)p(\theta)\int_{\underline{z}}^{\infty} g(z')S(z')dz'.$$

The employment density evolves according to the Kolmogorov forward equation (18), together with the cutoff destruction rule.

Evaluating these equations at the steady state, the steady-state equilibrium can be characterized by two equations with two unknowns, which are useful later.

Lemma 1 *The steady-state equilibrium market tightness and reservation match quality, $\{\theta, \underline{z}\}$, jointly solve*

$$0 = A\underline{z} - h - \frac{A(1-\zeta)}{\zeta}\int_{\underline{z}}^{\infty}\left[1 - \left(\frac{r+\sigma}{\Gamma(\tilde{z};\theta)}\right)^{\frac{\gamma}{1-\omega(1-\gamma)}}\right]d\tilde{z} \tag{51}$$

and

$$\begin{aligned}\kappa = (1-\gamma)q(\theta)\Bigg[&A\int_{\underline{z}}^{\infty}\frac{\sigma(1-G(z))}{\sigma+(1-G(z))\zeta p(\theta)}\frac{1}{\Gamma(z,\theta)}\,dz \\ &+(1-\omega)\int_{\underline{z}}^{\infty}\frac{\sigma g(z)\zeta p(\theta)(1-G(z))}{\left(\sigma+(1-G(z))\zeta p(\theta)\right)^2}\left(1+\frac{(1-\gamma)\left(\sigma+(1-G(z))\zeta p(\theta)\right)}{\Gamma(z,\theta)}\right)S(z)\,dz\Bigg],\end{aligned} \tag{52}$$

where $\Gamma(\tilde{z};\theta) \equiv r+\sigma+(1-\omega(1-\gamma))\zeta p(\theta)(1-G(\tilde{z}))$ and

$$S(z) = A\int_{\underline{z}}^{z}\Gamma(z,\theta)^{\frac{(\omega-1)(1-\gamma)}{1-\omega(1-\gamma)}}/\Gamma(\tilde{z},\theta)^{\frac{\gamma}{1-\omega(1-\gamma)}}\,d\tilde{z}$$

for $z \geq \underline{z}$. Moreover, (51) defines a weakly increasing relationship between $\underline{z}$ and θ, which we write as $\underline{z}^R(\theta)$, and (52) defines a strictly decreasing relationship between $\underline{z}$ and θ, which we write as $\underline{z}^{FE}(\theta)$.

Similar characterizations obtain for the planner's solution. Evaluating (21) in the steady state and taking a derivative with respect to z, we obtain

$$\left[r+\sigma+\zeta p(\theta)(1-G(z))\right]\mu'(z) = A,$$

with a boundary condition $\mu(\underline{z}) = 0$. The reservation match quality $\underline{z}$ solves

$$0 = A\underline{z} - h - (1-\zeta)p(\theta)\int_{\underline{z}}^{\infty}g(z')\mu(z')dz' + \kappa\theta(1-\zeta).$$

Evaluating these equations at the steady state, the efficient steady-state allocation can be tractably characterized as follows.

Lemma 2 *The efficient steady-state market tightness and reservation match quality* $\{\theta^{SP}, \underline{z}^{SP}\}$ *jointly solve*

$$0 = A\underline{z}^{SP} - h - A(1-\zeta)p(\theta^{SP})\int_{\underline{z}^{SP}}^{\infty} \frac{\sigma\eta(\theta^{SP}) + (1-G(\tilde{z}))\zeta p(\theta^{SP})}{\sigma + (1-G(\tilde{z}))\zeta p(\theta^{SP})} \frac{1-G(\tilde{z})}{r+\sigma+\zeta p(\theta^{SP})(1-G(\tilde{z}))} d\tilde{z} \quad (53)$$

and

$$\kappa = A(1-\eta(\theta^{SP}))q(\theta^{SP})\int_{\underline{z}^{SP}}^{\infty} \frac{\sigma}{\sigma + \zeta p(\theta^{SP})(1-G(\tilde{z}))} \frac{1-G(\tilde{z})}{r+\sigma+\zeta p(\theta^{SP})(1-G(\tilde{z}))} d\tilde{z}. \quad (54)$$

Moreover, if $\eta'(\theta) > -\eta(\theta)(1-\eta(\theta))/\theta$ *for all* $\theta > 0$, (53) *defines a weakly increasing relationship between* $\underline{z}^{SP}$ *and* θ*, which we write as* $\underline{z}^{R,SP}(\theta)$*, and* (54) *defines a strictly decreasing relationship between* $\underline{z}^{SP}$ *and* θ*, which we write as* $\underline{z}^{FE,SP}(\theta)$*.*

D Implementation with simple instruments

We consider a simple implementation of the efficient allocation using two instruments: a vacancy tax and an unemployment insurance benefit. The government imposes a vacancy tax χ_t^e on firms so that the total cost of posting a vacancy is $\kappa + \chi_t^e$ in the free-entry condition (17). The government also imposes an unemployment insurance benefit, b_t, on unemployed workers so that the flow value of unemployment is $h + b_t$. The government finances these fiscal instruments through a lump-sum tax or subsidy on workers. The unemployment insurance benefit replaces h with $h + b_t$ in the joint match surplus (15). The rest of the equilibrium conditions remain unchanged.

Proposition 5 *There exists a vacancy tax χ^e and an unemployment insurance benefit b that implement the efficient allocation in the steady state. When the Hosios condition $\gamma = \eta(\theta^{SP})$ holds and in the presence of on-the-job search ($\zeta > 0$), the vacancy tax is positive and the unemployment insurance benefit is nonnegative, that is, $\chi^e > 0$ and $b \geq 0$ (with strict inequality if $\zeta < 1$).*

Because there are only two endogenous variables (the reservation match quality $\underline{z}$ and the market tightness θ), these two instruments are sufficient to implement the efficient allocation, as the first part of Proposition 5 demonstrates. The second part of the proposition shows that, under the Hosios condition, the vacancy tax is positive and the unemployment insurance benefit is nonnegative whenever there is on-the-job search. As we show in Proposition 1, the equilibrium market tightness is higher than the efficient level. The positive vacancy tax corrects excessive job creation. Moreover, as the decentralized equilibrium tends to overvalue a job, the positive unemployment insurance benefit raises the reservation match quality, thereby preventing the formation of excessively unproductive jobs.

D.1 Quantitative magnitudes

We quantify the magnitudes of the optimal unemployment insurance and the vacancy tax using our baseline calibration in Section 5. Table D.3 reports the unemployment insurance and the vacancy tax that implement the efficient allocation under the baseline calibration of Section 5. Both are quantitatively substantial: the optimal unemployment insurance is 4.3% of the average output per worker, and the vacancy tax is 22.0% of the cost of vacancy creation. Both are positive, in line with Proposition 5, as we impose the Hosios condition.

	Calibration	Efficient
Unemployment Insurance (% of output per worker)	0.0%	4.3%
Entry tax (% of vacancy cost κ)	0.0%	22.0%

Table D.3: Efficient Unemployment Insurance and Vacancy Tax

Note: The table shows the unemployment insurance and the vacancy tax that achieve the efficient allocation under the baseline calibration.

E Proofs

E.1 Proof of Lemma 1

In the steady state, the match surplus $S(z)$ satisfies

$$\begin{aligned}(r+\sigma)S(z) = Az - h \\ + \zeta p(\theta)\int_z^\infty g(z')\left((\omega-1)S(z) + \gamma\left[S(z') - \omega S(z)\right]\right)dz' \\ - p(\theta)\int_{\underline{z}}^\infty g(z')\gamma S(z')dz'.\end{aligned} \tag{55}$$

Taking a derivative with respect to z, we have

$$\left[r+\sigma+(1-\omega(1-\gamma))\zeta p(\theta)(1-G(z))\right]S'(z) + \zeta p(\theta)g(z)(\omega-1)(1-\gamma)S(z) = A.$$

With a boundary condition $S(\underline{z}) = 0$, solving this ODE, we obtain

$$S(z;\underline{z},\theta) = \begin{cases} A\int_{\underline{z}}^z \frac{\left[r+\sigma+(1-\omega(1-\gamma))\zeta p(\theta)(1-G(z))\right]^{\frac{(\omega-1)(1-\gamma)}{(1-\omega(1-\gamma))}}}{\left[r+\sigma+(1-\omega(1-\gamma))\zeta p(\theta)(1-G(\tilde{z}))\right]^{\frac{\gamma}{(1-\omega(1-\gamma))}}}d\tilde{z}. & \text{for} \quad z \geq \underline{z} \\ 0 & \text{for} \quad z < \underline{z}. \end{cases} \tag{56}$$

Evaluating (55) at $z = \underline{z}$ yields

$$0 = A\underline{z} - h - \gamma(1-\zeta)p(\theta)\int_{\underline{z}}^\infty g(z')S(z';\underline{z},\theta)dz'.$$

Changing the order of integration and using (56), we have

$$\int_{\underline{z}}^{\infty} g(z')S(z';\underline{z},\theta)dz' = A\int_{\underline{z}}^{\infty} \frac{1}{\Gamma(\tilde{z};\theta)^{\frac{\gamma}{1-\omega(1-\gamma)}}} \left[\int_{\tilde{z}}^{\infty} g(z')\Gamma(z';\theta)^{\frac{(\omega-1)(1-\gamma)}{1-\omega(1-\gamma)}} dz'\right] d\tilde{z}$$
$$= \frac{A}{\gamma\zeta p(\theta)}\int_{\underline{z}}^{\infty}\left[1-\left(\frac{r+\sigma}{\Gamma(\tilde{z};\theta)}\right)^{\frac{\gamma}{1-\omega(1-\gamma)}}\right] d\tilde{z},$$

where

$$\Gamma(z;\theta) \equiv r+\sigma+(1-\omega(1-\gamma))\zeta p(\theta)(1-G(z)).$$

The second equality uses $\partial_z\Gamma(z;\theta) = -(1-\omega(1-\gamma))\zeta p(\theta)g(z)$. When $\gamma\zeta p(\theta) = 0$, the integral formula is interpreted by continuity. Plugging it back, we have

$$0 = A\underline{z} - h - \frac{A(1-\zeta)}{\zeta}\int_{\underline{z}}^{\infty}\left[1-\left(\frac{r+\sigma}{\Gamma(\tilde{z};\theta)}\right)^{\frac{\gamma}{1-\omega(1-\gamma)}}\right] d\tilde{z}$$
$$= H^R(\underline{z},\theta).$$

The solution to the above equation can be used to define the mapping $\underline{z}^R(\theta)$. Its derivative is

$$\frac{d\underline{z}^R(\theta)}{d\theta} = -\frac{\frac{\partial H^R(\underline{z},\theta)}{\partial\theta}}{\frac{\partial H^R(\underline{z},\theta)}{\partial\underline{z}}}.$$

The denominator is positive, that is, $\partial H^R(\underline{z},\theta)/\partial\underline{z} > 0$. For $\gamma > 0$, differentiating the expression for H^R gives

$$\frac{\partial H^R(\underline{z},\theta)}{\partial\theta} = -A\gamma(1-\zeta)p'(\theta)\int_{\underline{z}}^{\infty}\frac{1-G(z)}{\Gamma(z;\theta)}\left(\frac{r+\sigma}{\Gamma(z;\theta)}\right)^{\frac{\gamma}{1-\omega(1-\gamma)}} dz \leq 0.$$

When $\gamma = 0$, $H^R(\underline{z},\theta) = A\underline{z} - h$, so its derivative with respect to θ is zero. Therefore, $\underline{z}^R(\theta)$ is weakly increasing.

Let $P(z)$ denote the mass of workers employed with match quality below z or unemployed, where $P(\underline{z})$ corresponds to the unemployment rate. Its law of motion in the steady state is

$$0 = -(1-G(z))p(\theta)P(\underline{z}) - (P(z)-P(\underline{z}))(1-G(z))\zeta p(\theta) + \sigma(1-P(z)).$$

Solving for $P(z)$, we have

$$P(z) = \begin{cases} \frac{\sigma-(1-\zeta)(1-G(z))p(\theta)P(\underline{z})}{\sigma+(1-G(z))\zeta p(\theta)} & \text{for} \quad z \geq \underline{z} \\ \frac{\sigma}{p(\theta)(1-G(\underline{z}))+\sigma} & \text{for} \quad z < \underline{z}. \end{cases}$$

Define the probability that a vacancy meets a worker who is either unemployed or already employed with match quality below z as

$$\begin{aligned} F^{\ell}(z) &\equiv \frac{P(\underline{z}) + \zeta(P(z)-P(\underline{z}))}{P(\underline{z}) + \zeta(1-P(\underline{z}))} \\ &= \begin{cases} \frac{\sigma}{\sigma+(1-G(z))\zeta p(\theta)} & \text{for} \quad z \geq \underline{z} \\ \frac{\sigma}{\sigma+(1-G(\underline{z}))\zeta p(\theta)} & \text{for} \quad z < \underline{z} \end{cases} \equiv F^{\ell}(z;\theta,\underline{z}). \end{aligned}$$

The continuous part of this distribution has density

$$f(z) = \frac{\sigma g(z)\zeta p(\theta)}{(\sigma+(1-G(z))\zeta p(\theta))^2}, \qquad z > \underline{z}.$$

There is also an atom at $z = 0$ of size $F^{\ell}(\underline{z};\theta,\underline{z})$, corresponding to unemployed workers.

Using $F^{\ell}(z;\theta,\underline{z})$, the free-entry condition in the steady state can be written as

$$\kappa = (1-\gamma)q(\theta)\int_0^{\infty}\int_{z'}^{\infty} g(z)\left[S(z;\underline{z},\theta) - \omega S(z';\underline{z},\theta)\right] dz dF^{\ell}(z';\theta,\underline{z}). \tag{57}$$

This equation defines a mapping $\underline{z}^{FE}(\theta)$.

When $\underline{z}$ changes, the change in the unemployment atom cancels the boundary term

from the employed-worker integral, since $S(\underline{z};\underline{z},\theta)=0$. Taking derivatives,

$$\frac{d\underline{z}^{FE}(\theta)}{d\theta}=-\frac{\frac{\partial}{\partial\theta}\left[q(\theta)\int_0^\infty\int_{z'}^\infty g(z)\left[S(z;\underline{z},\theta)-\omega S(z';\underline{z},\theta)\right]dzdF^\ell(z';\theta,\underline{z})\right]}{q(\theta)\int_0^\infty\int_{z'}^\infty g(z)\left[\frac{\partial S(z;\underline{z},\theta)}{\partial\underline{z}}-\omega\frac{\partial S(z';\underline{z},\theta)}{\partial\underline{z}}\right]dzdF^\ell(z';\theta,\underline{z})}. \tag{58}$$

We would like to sign the above object. Note the denominator of (58) is negative because

$$\frac{\partial^2 S(z;\underline{z},\theta)}{\partial z\partial\underline{z}}=-A(1-\omega)(1-\gamma)\zeta p(\theta)g(z)\frac{[r+\sigma+(1-\omega(1-\gamma))\zeta p(\theta)(1-G(z))]^{\frac{(\omega-1)(1-\gamma)}{(1-\omega(1-\gamma))}-1}}{[r+\sigma+(1-\omega(1-\gamma))\zeta p(\theta)(1-G(\underline{z}))]^{\frac{\gamma}{(1-\omega(1-\gamma))}}}$$
$$\leq 0,$$

and as a result,

$$q(\theta)\int_0^\infty\int_{z'}^\infty g(z)\left[\frac{\partial S(z;\underline{z},\theta)}{\partial\underline{z}}-\omega\frac{\partial S(z';\underline{z},\theta)}{\partial\underline{z}}\right]dzdF^\ell(z';\theta,\underline{z})$$
$$\leq q(\theta)F^\ell(\underline{z};\theta,\underline{z})\int_0^\infty g(z)\frac{\partial S(z;\underline{z},\theta)}{\partial\underline{z}}dz<0.$$

To sign the numerator of (58), first note that

$$\frac{\partial S(z;\underline{z},\theta)}{\partial\theta}=B_1(z)+AB_2(z),$$

where

$$B_1(z)=-\frac{(1-\omega)(1-\gamma)\zeta p'(\theta)(1-G(z))}{\Gamma(z;\theta)}S(z)\leq 0$$

and

$$B_2(z)=-\gamma\zeta p'(\theta)\int_{\underline{z}}^z\frac{1-G(\tilde{z})}{\Gamma(\tilde{z};\theta)}\frac{\Gamma(z;\theta)^{\frac{(\omega-1)(1-\gamma)}{1-\omega(1-\gamma)}}}{\Gamma(\tilde{z};\theta)^{\frac{\gamma}{1-\omega(1-\gamma)}}}d\tilde{z}\leq 0.$$

Thus $\partial_\theta S(z;\underline{z},\theta)\leq 0$. When $\gamma=0$ and $\omega=1$, the same conclusion follows directly from

$S(z) = A(z - \underline{z})/(r + \sigma)$. Including the unemployment atom and changing the order of integration, the integral in the free-entry condition is

$$\begin{aligned}
&\int_0^\infty \int_{z'}^\infty g(z)[S(z) - \omega S(z')]dz\, dF^\ell(z') \\
&\quad = \int_{\underline{z}}^\infty [F^\ell(z)g(z) - \omega(1 - G(z))f(z)]S(z)dz \\
&\quad = \int_{\underline{z}}^\infty \frac{\sigma g(z)[\sigma + (1-\omega)\zeta p(\theta)(1 - G(z))]}{[\sigma + \zeta p(\theta)(1 - G(z))]^2} S(z)dz.
\end{aligned}$$

The coefficient multiplying $S(z)$ is nonnegative and weakly decreasing in θ, since

$$\begin{aligned}
&\frac{\partial}{\partial \theta}\left\{\frac{\sigma g(z)[\sigma + (1-\omega)\zeta p(\theta)(1 - G(z))]}{[\sigma + \zeta p(\theta)(1 - G(z))]^2}\right\} \\
&\quad = -\frac{\sigma g(z)\zeta p'(\theta)(1 - G(z))[(1+\omega)\sigma + (1-\omega)\zeta p(\theta)(1 - G(z))]}{[\sigma + \zeta p(\theta)(1 - G(z))]^3} \leq 0.
\end{aligned}$$

Together with $\partial_\theta S \leq 0$ and $q'(\theta) < 0$, this inequality shows that the numerator of (58) is strictly negative wherever the free-entry condition holds, because $\kappa > 0$ implies that the free-entry integral is positive. Putting it all together, we have

$$\frac{d\underline{z}^{FE}(\theta)}{d\theta} < 0.$$

Now, we seek to rewrite (57). We can rewrite the integral as follows:

$$\int_0^\infty \int_{z'}^\infty g(z)\left[S(z) - \omega S(z')\right] dz dF^\ell(z') = \int_{\underline{z}}^\infty F^\ell(z)g(z)S(z)dz - \omega \int_{\underline{z}}^\infty (1 - G(z))S(z)f(z)dz.$$

The surplus ODE implies

$$S'(z) = \frac{A}{\Gamma(z;\theta)} + \frac{(1-\omega)(1-\gamma)\zeta p(\theta)g(z)S(z)}{\Gamma(z;\theta)}.$$

Using integration by parts, with the boundary term vanishing because $S(\underline{z}) = 0$, $S(z) \leq$

$A(z-\underline{z})/(r+\sigma)$, and G has a finite mean,

$$\begin{aligned}\int_{\underline{z}}^{\infty} F^{\ell}(z)g(z)S(z)dz &= -\left[F^{\ell}(z)(1-G(z))S(z)\right]_{\underline{z}}^{\infty} + \int_{\underline{z}}^{\infty} f(z)(1-G(z))S(z)dz \\ &\quad + \int_{\underline{z}}^{\infty} F^{\ell}(z)(1-G(z))S'(z)dz \\ = A\int_{\underline{z}}^{\infty} f(z)(1-G(z))\int_{\underline{z}}^{z} &\frac{\Gamma(z,\theta)^{\frac{(\omega-1)(1-\gamma)}{(1-\omega(1-\gamma))}}}{\Gamma(\tilde{z},\theta)^{\frac{\gamma}{(1-\omega(1-\gamma))}}}d\tilde{z}dz + A\int_{\underline{z}}^{\infty} F^{\ell}(z)(1-G(z))\frac{1}{\Gamma(z,\theta)}dz \\ &\quad +(1-\omega)(1-\gamma)\zeta p(\theta)\int_{\underline{z}}^{\infty}\frac{F^{\ell}(z)(1-G(z))g(z)S(z)}{\Gamma(z;\theta)}dz.\end{aligned}$$

Substituting the above expressions back and using the expressions for $F^{\ell}(z)$ and $f(z)$, we have

$$\begin{aligned}\int_0^{\infty}\int_{z'}^{\infty} g(z)\left[S(z)-\omega S(z')\right]dzdF^{\ell}(z') &= A\int_{\underline{z}}^{\infty}\frac{\sigma(1-G(z))}{\sigma+(1-G(z))\zeta p(\theta)}\frac{1}{\Gamma(z,\theta)}dz \\ &+ A(1-\omega)\int_{\underline{z}}^{\infty}\frac{\sigma g(z)\zeta p(\theta)(1-G(z))}{(\sigma+(1-G(z))\zeta p(\theta))^2}\left[1+\frac{(1-\gamma)(\sigma+(1-G(z))\zeta p(\theta))}{\Gamma(z,\theta)}\right] \\ &\times\int_{\underline{z}}^{z}\frac{\Gamma(z,\theta)^{\frac{(\omega-1)(1-\gamma)}{(1-\omega(1-\gamma))}}}{\Gamma(\tilde{z},\theta)^{\frac{\gamma}{(1-\omega(1-\gamma))}}}d\tilde{z}dz.\end{aligned}$$

Now, (57) can be rewritten as follows:

$$\begin{aligned}\kappa = (1-\gamma)q(\theta)A\Bigg[&\int_{\underline{z}}^{\infty}\frac{\sigma(1-G(z))}{\sigma+(1-G(z))\zeta p(\theta)}\frac{1}{\Gamma(z,\theta)}dz \\ &+(1-\omega)\int_{\underline{z}}^{\infty}\frac{\sigma g(z)\zeta p(\theta)(1-G(z))}{(\sigma+(1-G(z))\zeta p(\theta))^2}\left[1+\frac{(1-\gamma)(\sigma+(1-G(z))\zeta p(\theta))}{\Gamma(z,\theta)}\right] \\ &\times\int_{\underline{z}}^{z}\frac{\Gamma(z,\theta)^{\frac{(\omega-1)(1-\gamma)}{(1-\omega(1-\gamma))}}}{\Gamma(\tilde{z},\theta)^{\frac{\gamma}{(1-\omega(1-\gamma))}}}d\tilde{z}dz\Bigg].\end{aligned}$$

E.2 Proof of Lemma 2

Note $F^{\ell}(z;\theta)$ can be characterized in the same way as Lemma 1. We omit superscript SP for notational simplicity.

We start from equation (28) evaluated at the steady state:

$$
\begin{aligned}
(r+\sigma)\mu(z) &= Az - h \\
&\quad - p(\theta)\int_{\underline{z}}^{\infty} g(z')\mu(z')dz' + \zeta p(\theta)\int_{z}^{\infty} g(z')(\mu(z') - \mu(z))dz' \\
&\quad + (1-\zeta)(1-\eta(\theta))p(\theta)\left[\int_0^{\infty}\int_{z'}^{\infty} g(\tilde{z})\left(\mu(\tilde{z}) - \mu(z')\right)d\tilde{z}dF^{\ell}(z';\theta,\underline{z})\right].
\end{aligned} \tag{59}
$$

Taking the derivative with respect to z,

$$
\left[r+\sigma+\zeta p(\theta)(1-G(z))\right]\mu'(z) = A.
$$

Solving for $\mu(z)$ with a boundary condition $\mu(\underline{z}) = 0$ gives

$$
\mu(z;\underline{z},\theta) = \begin{cases} A\int_{\underline{z}}^{z} \frac{1}{r+\sigma+\zeta p(\theta)(1-G(\tilde{z}))}d\tilde{z}. & \text{for} \quad z \geq \underline{z} \\ 0 & \text{for} \quad z < \underline{z} \end{cases}
$$

Evaluating (59) at $z = \underline{z}$ gives

$$
\begin{aligned}
0 &= A\underline{z} - h \\
&\quad - (1-\zeta)p(\theta)\int_{\underline{z}}^{\infty} g(z')\mu(z';\underline{z},\theta)dz' \\
&\quad + (1-\zeta)(1-\eta(\theta))p(\theta)\left[\int_0^{\infty}\int_{z'}^{\infty} g(\tilde{z})\left(\mu(\tilde{z};\underline{z},\theta) - \mu(z';\underline{z},\theta)\right)d\tilde{z}dF^{\ell}(z';\theta,\underline{z})\right].
\end{aligned} \tag{60}
$$

Now, we seek to simplify the above expression. We rewrite the second line using integration by parts:

$$
\int_{\underline{z}}^{\infty} g(z')\mu(z';\underline{z},\theta)dz' = A\int_{\underline{z}}^{\infty}(1-G(\tilde{z}))\frac{1}{r+\sigma+\zeta p(\theta)(1-G(\tilde{z}))}d\tilde{z}, \tag{61}
$$

We rewrite the last term as

$$\begin{aligned}
&\int_0^\infty \int_{z'}^\infty g(\tilde{z}) \left(\mu(\tilde{z}) - \mu(z';\underline{z},\theta)\right) d\tilde{z} dF^\ell(z') \\
&= \int_{\underline{z}}^\infty F^\ell(\tilde{z}) g(\tilde{z}) \mu(\tilde{z};\underline{z},\theta) d\tilde{z} - \int_{\underline{z}}^\infty (1 - G(z')) \mu(z';\underline{z},\theta) f(z') dz' \\
&= A \int_{\underline{z}}^\infty F^\ell(\tilde{z})(1 - G(\tilde{z})) \frac{1}{r + \sigma + \zeta p(\theta)(1 - G(\tilde{z}))} d\tilde{z} \\
&= A \int_{\underline{z}}^\infty \frac{\sigma}{\sigma + \zeta p(\theta)(1 - G(\tilde{z}))} \frac{1 - G(\tilde{z})}{r + \sigma + \zeta p(\theta)(1 - G(\tilde{z}))} d\tilde{z}, \qquad (62)
\end{aligned}$$

where the second line changes the order of integration, and the third line uses integration by parts. Plugging (61) and (62) into (60), we obtain

$$\begin{aligned}
0 &= A\underline{z} - h - A(1 - \zeta) p(\theta) \int_{\underline{z}}^\infty \frac{\sigma\eta(\theta) + (1 - G(\tilde{z}))\zeta p(\theta)}{\sigma + (1 - G(\tilde{z}))\zeta p(\theta)} \frac{1 - G(\tilde{z})}{r + \sigma + \zeta p(\theta)(1 - G(\tilde{z}))} d\tilde{z} \\
&\equiv H^R(\underline{z}, \theta).
\end{aligned}$$

The solution to the above equation gives a mapping $\underline{z}^R(\theta)$, and its derivative is given by

$$\frac{d\underline{z}^R(\theta)}{d\theta} = -\frac{\frac{\partial H^R(\underline{z},\theta)}{\partial \theta}}{\frac{\partial H^R(\underline{z},\theta)}{\partial \underline{z}}}.$$

Clearly, the denominator is positive, that is, $\frac{\partial H^R(\underline{z},\theta)}{\partial \underline{z}} > 0$. Using $p'(\theta)/p(\theta) = (1 - \eta(\theta))/\theta$, we obtain

$$\begin{aligned}
\frac{\partial H^R(\underline{z},\theta)}{\partial \theta} = -A(1 - \zeta) p(\theta) \int_{\underline{z}}^\infty & \frac{1 - G(\tilde{z})}{(\sigma + (1 - G(\tilde{z}))\zeta p(\theta))^2 (r + \sigma + \zeta p(\theta)(1 - G(\tilde{z})))^2} \\
& \times \Bigg[\sigma \frac{p'(\theta)}{p(\theta)} \left(\sigma\eta(\theta) + (1 - G(\tilde{z}))\zeta p(\theta)\right) \left(r + \sigma + \zeta p(\theta)(1 - G(\tilde{z}))\right) \\
& + \sigma\eta'(\theta) \left(\sigma + (1 - G(\tilde{z}))\zeta p(\theta)\right) \left(r + \sigma + \zeta p(\theta)(1 - G(\tilde{z}))\right) \\
& + (1 - G(\tilde{z}))\zeta p'(\theta) \left(\sigma + (1 - G(\tilde{z}))\zeta p(\theta)\right) \left(r + \sigma(1 - \eta(\theta))\right) \Bigg] d\tilde{z}.
\end{aligned}$$

This expression is weakly negative if $\eta'(\theta) \geq k^R$, where

$$\begin{aligned} k^R \equiv \sup_{\tilde{z} \geq \underline{z}} \Bigg\{ & -\frac{p'(\theta)}{p(\theta)} \frac{\sigma\eta(\theta) + (1 - G(\tilde{z}))\zeta p(\theta)}{\sigma + (1 - G(\tilde{z}))\zeta p(\theta)} \\ & - \frac{(1 - G(\tilde{z}))\zeta p'(\theta)(r + \sigma(1 - \eta(\theta)))}{\sigma(r + \sigma + \zeta p(\theta)(1 - G(\tilde{z})))} \Bigg\} \\ = & -\frac{\eta(\theta)p'(\theta)}{p(\theta)} < 0. \end{aligned}$$

The last equality follows because the expression inside braces is at most $-\eta(\theta)p'(\theta)/p(\theta)$ and converges to that value as $\tilde{z} \to \infty$. Therefore, $\underline{z}^R(\theta)$ is weakly increasing if $\eta'(\theta) \geq k^R$.

We can rewrite (23) to obtain:

$$\kappa = (1 - \eta(\theta))q(\theta) \left[\int_0^\infty \int_{z'}^\infty g(\tilde{z}) \left(\mu(\tilde{z}; \underline{z}, \theta) - \mu(z'; \underline{z}, \theta) \right) d\tilde{z} dF^\ell(z'; \theta, \underline{z}) \right].$$

Substituting (62), we have

$$\begin{aligned} \kappa &= A(1 - \eta(\theta))q(\theta) \int_{\underline{z}}^\infty \frac{\sigma}{\sigma + \zeta p(\theta)(1 - G(\tilde{z}))} \frac{1 - G(\tilde{z})}{r + \sigma + \zeta p(\theta)(1 - G(\tilde{z}))} d\tilde{z} \\ &\equiv H^{FE}(\underline{z}, \theta). \end{aligned}$$

This equation defines a mapping $\underline{z}^{FE}(\theta)$. To determine the sign of $d\underline{z}^{FE}(\theta)/d\theta$, note that

$$\frac{d\underline{z}^{FE}(\theta)}{d\theta} = -\frac{\frac{\partial H^{FE}(\underline{z},\theta)}{\partial \theta}}{\frac{\partial H^{FE}(\underline{z},\theta)}{\partial \underline{z}}}.$$

Clearly, $\frac{\partial H^{FE}(\underline{z},\theta)}{\partial \underline{z}} < 0$, and the denominator is negative. To sign the numerator,

$$\begin{aligned}
&\frac{\partial H^{FE}(\underline{z},\theta)}{\partial \theta} \\
&= -A\eta'(\theta)q(\theta)\int_{\underline{z}}^{\infty} \frac{\sigma}{\sigma+\zeta p(\theta)(1-G(\tilde{z}))}\frac{1-G(\tilde{z})}{r+\sigma+\zeta p(\theta)(1-G(\tilde{z}))}d\tilde{z} \\
&\quad + A(1-\eta(\theta))q'(\theta)\int_{\underline{z}}^{\infty} \frac{\sigma}{\sigma+\zeta p(\theta)(1-G(\tilde{z}))}\frac{1-G(\tilde{z})}{r+\sigma+\zeta p(\theta)(1-G(\tilde{z}))}d\tilde{z} \\
&\quad - A(1-\eta(\theta))q(\theta)\int_{\underline{z}}^{\infty} \frac{\sigma(1-G(\tilde{z}))\zeta p'(\theta)(1-G(\tilde{z}))}{(\sigma+\zeta p(\theta)(1-G(\tilde{z})))}\frac{1}{(r+\sigma+\zeta p(\theta)(1-G(\tilde{z})))} \\
&\qquad \times\left(\frac{1}{\sigma+\zeta p(\theta)(1-G(\tilde{z}))}+\frac{1}{r+\sigma+\zeta p(\theta)(1-G(\tilde{z}))}\right)d\tilde{z}.
\end{aligned}$$

This expression is strictly negative if $\eta'(\theta) > k^{FE}$, where

$$\begin{aligned}
k^{FE} \equiv &-\frac{(1-\eta(\theta))\eta(\theta)}{\theta} \\
&-\frac{1}{\int_{\underline{z}}^{\infty}\frac{\sigma}{\sigma+\zeta p(\theta)(1-G(\tilde{z}))}\frac{1-G(\tilde{z})}{r+\sigma+\zeta p(\theta)(1-G(\tilde{z}))}d\tilde{z}}\times \\
&(1-\eta(\theta))\int_{\underline{z}}^{\infty}\frac{\sigma(1-G(\tilde{z}))\zeta p'(\theta)(1-G(\tilde{z}))}{(\sigma+\zeta p(\theta)(1-G(\tilde{z})))(r+\sigma+\zeta p(\theta)(1-G(\tilde{z})))} \\
&\qquad\times\left(\frac{1}{\sigma+\zeta p(\theta)(1-G(\tilde{z}))}+\frac{1}{r+\sigma+\zeta p(\theta)(1-G(\tilde{z}))}\right)d\tilde{z} \\
&< 0.
\end{aligned}$$

Therefore, if $\eta'(\theta) > k^{FE}$, $\underline{z}^{FE}(\theta)$ is strictly decreasing. Since $k^{FE} \leq k^R = -\eta(\theta)(1-\eta(\theta))/\theta$, the condition in Lemma 2 implies both slope conclusions, completing the proof.

E.3 Proof of Proposition 1

We state and prove a more general version of Proposition 1.

Proposition 6 *There exists $\hat{\gamma} > \eta(\theta^{SP})$ such that for $\gamma < \hat{\gamma}$, the steady-state equilibrium market tightness θ in the decentralized equilibrium is higher than the socially efficient level when $\zeta > 0$.*

Proof. Fix θ with $\gamma \leq \eta(\theta)$. We first show $\underline{z}^R(\theta) \leq \underline{z}^{R,SP}(\theta)$. Let

$$H^R(\underline{z},\theta) \equiv A\underline{z} - h - \frac{A(1-\zeta)}{\zeta}\int_{\underline{z}}^{\infty}\left[1-\left(\frac{r+\sigma}{\Gamma(\tilde{z};\theta)}\right)^{\frac{\gamma}{1-\omega(1-\gamma)}}\right]d\tilde{z}$$

and

$$H^{R,SP}(\underline{z},\theta) \equiv A\underline{z} - h - A(1-\zeta)p(\theta)\int_{\underline{z}}^{\infty}\frac{\sigma\eta(\theta)+(1-G(\tilde{z}))\zeta p(\theta)}{\sigma+(1-G(\tilde{z}))\zeta p(\theta)}\frac{1-G(\tilde{z})}{r+\sigma+\zeta p(\theta)(1-G(\tilde{z}))}d\tilde{z}.$$

Because $\underline{z}^R(\theta)$ and $\underline{z}^{R,SP}(\theta)$ are the solutions to $H^R(\underline{z},\theta) = 0$ and $H^{R,SP}(\underline{z},\theta) = 0$, and both residuals are strictly increasing in $\underline{z}$, $H^R(\underline{z},\theta) \geq H^{R,SP}(\underline{z},\theta)$ is sufficient for $\underline{z}^R(\theta) \leq \underline{z}^{R,SP}(\theta)$. The residual inequality holds if

$$\frac{\sigma\eta(\theta)+(1-G(\tilde{z}))\zeta p(\theta)}{\sigma+(1-G(\tilde{z}))\zeta p(\theta)}\frac{1-G(\tilde{z})}{r+\sigma+\zeta p(\theta)(1-G(\tilde{z}))} - \frac{1}{\zeta p(\theta)}\left[1-\left(\frac{r+\sigma}{\Gamma(\tilde{z};\theta)}\right)^{\frac{\gamma}{1-\omega(1-\gamma)}}\right] \geq 0$$

for all $\tilde{z}$.

To show the above inequality,

$$\begin{aligned}
&\frac{\sigma\eta(\theta)+(1-G(\tilde{z}))\zeta p(\theta)}{\sigma+(1-G(\tilde{z}))\zeta p(\theta)}\frac{1-G(\tilde{z})}{r+\sigma+\zeta p(\theta)(1-G(\tilde{z}))} - \frac{1}{\zeta p(\theta)}\left[1-\left(\frac{r+\sigma}{\Gamma(\tilde{z};\theta)}\right)^{\frac{\gamma}{1-\omega(1-\gamma)}}\right] \\
&\geq (1-G(\tilde{z}))\left(\frac{\sigma\eta(\theta)+(1-G(\tilde{z}))\zeta p(\theta)}{\sigma+(1-G(\tilde{z}))\zeta p(\theta)}\frac{1}{r+\sigma+\zeta p(\theta)(1-G(\tilde{z}))} - \frac{\gamma}{r+\sigma+\gamma\zeta p(\theta)(1-G(\tilde{z}))}\right) \\
&= (1-G(\tilde{z}))\,\frac{(\eta(\theta)-\gamma)\sigma\left(r+\sigma+\gamma\zeta p(\theta)(1-G(\tilde{z}))\right)+\zeta p(\theta)(1-G(\tilde{z}))(1-\gamma)\left[r+\sigma(1-\gamma)\right]}{(\sigma+\zeta p(\theta)(1-G(\tilde{z})))(r+\sigma+\zeta p(\theta)(1-G(\tilde{z})))(r+\sigma+\gamma\zeta p(\theta)(1-G(\tilde{z})))},
\end{aligned} \tag{63}$$

The inequality follows from

$$\begin{aligned}\left(\frac{r+\sigma}{\Gamma(\tilde{z};\theta)}\right)^{\frac{\gamma}{1-\omega(1-\gamma)}} &= \left(\frac{r+\sigma}{r+\sigma+(1-\omega(1-\gamma))\zeta p(\theta)(1-G(\tilde{z}))}\right)^{\frac{\gamma}{1-\omega(1-\gamma)}} \\ &= \frac{1}{\left(1+\frac{1}{r+\sigma}(1-\omega(1-\gamma))\zeta p(\theta)(1-G(\tilde{z}))\right)^{\frac{\gamma}{1-\omega(1-\gamma)}}} \\ &\geq \frac{1}{\left(1+\frac{\gamma}{1-\omega(1-\gamma)}\frac{1}{r+\sigma}(1-\omega(1-\gamma))\zeta p(\theta)(1-G(\tilde{z}))\right)} \\ &= 1-\frac{\gamma\zeta p(\theta)(1-G(\tilde{z}))}{r+\sigma+\gamma\zeta p(\theta)(1-G(\tilde{z}))},\end{aligned}$$

where the third line follows from the fact that $(1+x)^{\tilde{n}} \leq 1+\tilde{n}x$ for $\tilde{n} \in [0,1]$ and $x \geq -1$, applied with $\tilde{n} \equiv \gamma/[1-\omega(1-\gamma)]$ and $x = (1-\omega(1-\gamma))\zeta p(\theta)(1-G(\tilde{z}))/(r+\sigma)$. The final expression (63) is nonnegative whenever $\eta(\theta) \geq \gamma$. Therefore $\underline{z}^R(\theta) \leq \underline{z}^{R,SP}(\theta)$ at such θ with strict inequality when $\zeta < 1$.

Next, at the same θ with $\gamma \leq \eta(\theta)$, we compare the free-entry conditions. The cutoffs $\underline{z}^{FE}(\theta)$ and $\underline{z}^{FE,SP}(\theta)$ solve $H^{FE}(\underline{z},\theta) = 0$ and $H^{FE,SP}(\underline{z},\theta) = 0$, respectively, where

$$\begin{aligned}H^{FE}(\underline{z},\theta) = A(1-\gamma)q(\theta)\Bigg[&\int_{\underline{z}}^{\infty}\frac{\sigma(1-G(z))}{\sigma+(1-G(z))\zeta p(\theta)}\frac{1}{\Gamma(z,\theta)}dz \\ &+(1-\omega)\int_{\underline{z}}^{\infty}\frac{\sigma g(z)\zeta p(\theta)(1-G(z))}{(\sigma+(1-G(z))\zeta p(\theta))^2}\left[1+\frac{(1-\gamma)(\sigma+(1-G(z))\zeta p(\theta))}{\Gamma(z,\theta)}\right] \\ &\times\int_{\underline{z}}^{z}\frac{\Gamma(z,\theta)^{\frac{(\omega-1)(1-\gamma)}{(1-\omega(1-\gamma))}}}{\Gamma(\tilde{z},\theta)^{\frac{\gamma}{(1-\omega(1-\gamma))}}}d\tilde{z}dz\Bigg]-\kappa\end{aligned}$$

and

$$H^{FE,SP}(\underline{z},\theta) = A(1-\eta(\theta))q(\theta)\int_{\underline{z}}^{\infty}\frac{\sigma}{\sigma+\zeta p(\theta)(1-G(\tilde{z}))}\frac{1-G(\tilde{z})}{r+\sigma+\zeta p(\theta)(1-G(\tilde{z}))}d\tilde{z}-\kappa.$$

Because $H^{FE}(\underline{z},\theta)$ and $H^{FE,SP}(\underline{z},\theta)$ are both strictly decreasing in $\underline{z}$, showing $H^{FE}(\underline{z},\theta) >$

$H^{FE,SP}(\underline{z},\theta)$ implies $\underline{z}^{FE}(\theta) > \underline{z}^{FE,SP}(\theta)$ whenever these cutoffs are defined. To show this inequality,

$$
\begin{aligned}
& H^{FE}(\underline{z},\theta) - H^{FE,SP}(\underline{z},\theta) \\
&= A(1-\gamma)q(\theta)\left[\int_{\underline{z}}^{\infty} \frac{\sigma(1-G(z))}{\sigma+(1-G(z))\zeta p(\theta)}\left(\frac{1}{\Gamma(z,\theta)} - \frac{1}{r+\sigma+\zeta p(\theta)(1-G(z))}\right)dz\right. \\
&\qquad + (1-\omega)\int_{\underline{z}}^{\infty} \frac{\sigma g(z)\zeta p(\theta)(1-G(z))}{(\sigma+(1-G(z))\zeta p(\theta))^2}\left[1+\frac{(1-\gamma)(\sigma+(1-G(z))\zeta p(\theta))}{\Gamma(z,\theta)}\right] \\
&\qquad \left.\times \int_{\underline{z}}^{z} \frac{\Gamma(z,\theta)^{\frac{(\omega-1)(1-\gamma)}{(1-\omega(1-\gamma))}}}{\Gamma(\tilde{z},\theta)^{\frac{\gamma}{(1-\omega(1-\gamma))}}} d\tilde{z}dz\right] \\
&\qquad + A(\eta(\theta)-\gamma)q(\theta)\int_{\underline{z}}^{\infty} \frac{\sigma}{\sigma+\zeta p(\theta)(1-G(\tilde{z}))}\frac{1-G(\tilde{z})}{r+\sigma+\zeta p(\theta)(1-G(\tilde{z}))}d\tilde{z} \\
&> 0.
\end{aligned}
$$

Now evaluate these comparisons at $\theta = \theta^{SP}$. For every $\gamma \leq \eta(\theta^{SP})$,

$$\underline{z}^{R}(\theta^{SP}) \leq \underline{z}^{SP} = \underline{z}^{FE,SP}(\theta^{SP}) < \underline{z}^{FE}(\theta^{SP}).$$

By Lemma 1, $\underline{z}^{R}(\theta)$ is weakly increasing and $\underline{z}^{FE}(\theta)$ is strictly decreasing, so their intersection must satisfy $\theta^{DE} > \theta^{SP}$.

At fixed θ^{SP}, both decentralized cutoffs are continuous in γ. This fact implies that there exists $\hat{\gamma} > \eta(\theta^{SP})$ such that $\theta^{DE} > \theta^{SP}$ for all $\gamma < \hat{\gamma}$. ■

E.4 Proof of Proposition 5

We consider an alternative implementation of the efficient allocation through unemployment insurance and a vacancy tax. Unemployment insurance provides b units of consumption goods for unemployed workers so that the flow value of being unemployed is now given by $h + b$. The vacancy tax is given by χ^e, and the vacancy-posting cost inclusive of the tax is $\chi^e + \kappa$. These two policy instruments are sufficient to implement the

efficient allocation. The following proposition is a generalization of Proposition 5.

Proposition 7 *There exists a pair of unemployment insurance b and vacancy tax χ^e that implement the efficient allocation in the steady state. Moreover, when $\zeta > 0$, there exists $\bar{\gamma} > \eta(\theta^{SP})$ such that for $\gamma < \bar{\gamma}$, $b \geq 0$ and $\chi^e > 0$, with $b > 0$ when $\zeta < 1$.*

Proof. In the steady state of the decentralized equilibrium with the above policy instruments, $(\underline{z}, \theta)$ solve

$$0 = A\underline{z} - h - b - \frac{A(1-\zeta)}{\zeta} \int_{\underline{z}}^{\infty} \left[1 - \left(\frac{r+\sigma}{\Gamma(\tilde{z};\theta)} \right)^{\frac{\gamma}{1-\omega(1-\gamma)}} \right] d\tilde{z}$$

and

$$\begin{aligned} \kappa + \chi^e = A(1-\gamma)q(\theta) \Bigg[& \int_{\underline{z}}^{\infty} \frac{\sigma(1-G(z))}{\sigma + (1-G(z))\zeta p(\theta)} \frac{1}{\Gamma(z,\theta)} dz \\ & + (1-\omega) \int_{\underline{z}}^{\infty} \frac{\sigma g(z) \zeta p(\theta)(1-G(z))}{(\sigma + (1-G(z))\zeta p(\theta))^2} \left[1 + \frac{(1-\gamma)(\sigma + (1-G(z))\zeta p(\theta))}{\Gamma(z,\theta)} \right] \\ & \times \int_{\underline{z}}^{z} \frac{\Gamma(z,\theta)^{\frac{(\omega-1)(1-\gamma)}{(1-\omega(1-\gamma))}}}{\Gamma(\tilde{z},\theta)^{\frac{\gamma}{(1-\omega(1-\gamma))}}} d\tilde{z} dz \Bigg]. \end{aligned}$$

We compare the above two conditions with those for the efficient allocation:

$$0 = A\underline{z}^{SP} - h - A(1-\zeta)p(\theta^{SP}) \int_{\underline{z}^{SP}}^{\infty} \frac{\sigma\eta(\theta^{SP}) + (1-G(\tilde{z}))\zeta p(\theta^{SP})}{\sigma + (1-G(\tilde{z}))\zeta p(\theta^{SP})} \frac{1-G(\tilde{z})}{r+\sigma+\zeta p(\theta^{SP})(1-G(\tilde{z}))} d\tilde{z}$$

and

$$\kappa = A(1-\eta(\theta^{SP}))q(\theta^{SP}) \int_{\underline{z}^{SP}}^{\infty} \frac{\sigma}{\sigma + \zeta p(\theta^{SP})(1-G(\tilde{z}))} \frac{1-G(\tilde{z})}{r+\sigma+\zeta p(\theta^{SP})(1-G(\tilde{z}))} d\tilde{z}.$$

Therefore, we can obtain the equivalence by setting

$$b = -\frac{A(1-\zeta)}{\zeta}\int_{\underline{z}^{SP}}^{\infty}\left[1-\left(\frac{r+\sigma}{\Gamma(\tilde{z};\theta^{SP})}\right)^{\frac{\gamma}{1-\omega(1-\gamma)}}\right]d\tilde{z}$$
$$+A(1-\zeta)p(\theta^{SP})\int_{\underline{z}^{SP}}^{\infty}\frac{\sigma\eta(\theta^{SP})+(1-G(\tilde{z}))\zeta p(\theta^{SP})}{\sigma+(1-G(\tilde{z}))\zeta p(\theta^{SP})}\frac{1-G(\tilde{z})}{r+\sigma+\zeta p(\theta^{SP})(1-G(\tilde{z}))}d\tilde{z}$$

and

$$\chi^{e} = A(1-\gamma)q(\theta^{SP})\left[\int_{\underline{z}^{SP}}^{\infty}\frac{\sigma(1-G(z))}{\sigma+(1-G(z))\zeta p(\theta^{SP})}\frac{1}{\Gamma(z,\theta^{SP})}dz\right.$$
$$+(1-\omega)\int_{\underline{z}^{SP}}^{\infty}\frac{\sigma g(z)\zeta p(\theta^{SP})(1-G(z))}{(\sigma+(1-G(z))\zeta p(\theta^{SP}))^{2}}\left[1+\frac{(1-\gamma)(\sigma+(1-G(z))\zeta p(\theta^{SP}))}{\Gamma(z,\theta^{SP})}\right.$$

$$\left.\left.\times\int_{\underline{z}^{SP}}^{z}\frac{\Gamma(z,\theta^{SP})^{\frac{(\omega-1)(1-\gamma)}{(1-\omega(1-\gamma))}}}{\Gamma(\tilde{z},\theta^{SP})^{\frac{\gamma}{(1-\omega(1-\gamma))}}}d\tilde{z}dz\right]\right.$$
$$-A(1-\eta(\theta^{SP}))q(\theta^{SP})\int_{\underline{z}^{SP}}^{\infty}\frac{\sigma}{\sigma+\zeta p(\theta^{SP})(1-G(\tilde{z}))}\frac{1-G(\tilde{z})}{r+\sigma+\zeta p(\theta^{SP})(1-G(\tilde{z}))}d\tilde{z}$$

The proof of Proposition 1 already shows that when $\zeta > 0$ and $\gamma \leq \eta(\theta^{SP})$,

$$b \geq 0, \quad \chi^{e} > 0.$$

For $0 < \zeta < 1$, the inequality for b is also strict, so continuity in γ yields $\bar{\gamma} > \eta(\theta^{SP})$ such that $b > 0$ and $\chi^{e} > 0$ for $\gamma < \bar{\gamma}$. When $\zeta = 1$, $b = 0$ for every γ, while continuity of the vacancy tax gives the same conclusion for χ^{e}. ■

E.5 Proof of Proposition 2

With the employment tax (33), the flow payoff in the joint surplus (15) becomes $A_t z - h - \tau(z)$. Substituting $\tau(z) = \Delta^{DE}(z, \theta_t^{SP})|_{S=\mu} - \Delta^{SP}(\theta_t^{SP})$ into the rewritten surplus equation (31) yields the planner's valuation equation (28) whenever $S_t = \mu_t$. With the vacancy tax (34), the free-entry condition (17) with vacancy cost $\kappa + \chi_t^e$ coincides with the planner's free-entry condition (23) at $S_t = \mu_t$. Hence the decentralized equilibrium implements the efficient allocation with $S_t(z) = \mu_t(z)$ for all z.

E.6 Proof of Proposition 3

Under $\gamma = \eta(\theta^{SP})$, the vacancy tax (34) reduces to

$$\chi_t^e = (1-\omega)(1-\gamma) q(\theta_t^{SP}) \int_{\underline{z}_t}^{\infty} \int_{z}^{\infty} f_t(z) g(z') \mu_t(z)\, dz' dz \geq 0,$$

with equality if and only if $\omega = 1$ (or there is no employed mass above the cutoff). This argument proves the second claim.

For the employment tax, write $\tau(z) = \Delta^{DE}(z, \theta^{SP})|_{S=\mu} - \Delta^{SP}(\theta^{SP})$. Differentiating (32) at $S = \mu$ gives

$$\partial_z \Delta^{DE}(z, \theta) = \zeta(1-\gamma) p(\theta) \Big[g(z)(1-\omega)\mu(z) + \omega \mu'(z)(1 - G(z)) \Big] \geq 0,$$

while Δ^{SP} does not depend on z. Hence τ is weakly increasing in z.

It remains to show $\tau(z) \geq 0$. Evaluating at the reservation cutoff and using $\mu(\underline{z}^{SP}) = 0$ together with $\gamma = \eta(\theta^{SP})$,

$$\tau(\underline{z}^{SP}) = (1-\zeta)(1-\gamma) p(\theta^{SP}) \Bigg[\int_{\underline{z}^{SP}}^{\infty} g(z') \mu(z')\, dz' - f^u \int_{\underline{z}^{SP}}^{\infty} g(z') \mu(z')\, dz' \\ - \int_{\underline{z}^{SP}}^{\infty} \int_{z}^{\infty} f(z) g(z') \big(\mu(z') - \mu(z) \big)\, dz' dz \Bigg].$$

Since $f^u + \int f(z)dz = 1$ from (6) and $\int_z^\infty g(z')(\mu(z') - \mu(z))\,dz' \leq \int_{\underline{z}^{SP}}^\infty g(z')\mu(z')\,dz'$, the expression in brackets is nonnegative. Thus $\tau(\underline{z}^{SP}) \geq 0$, and monotonicity yields $\tau(z) \geq 0$ for all $z \geq \underline{z}^{SP}$. When $\zeta = 1$, $\Delta^{SP} = 0$ and $\Delta^{DE}(\underline{z}^{SP}) = 0$, so $\tau(\underline{z}^{SP}) = 0$.

F Transition Dynamics in Sequence Space

Consider the deterministic transition following the productivity shock in Section 5.4. Take the productivity path and the initial employment distribution $n_0(z)$ as given. Write $\mathbf{A} = \{A_t\}_{t\geq 0}$, $\boldsymbol{\theta} = \{\theta_t\}_{t\geq 0}$, and $\underline{\mathbf{z}} = \{\underline{z}_t\}_{t\geq 0}$. Throughout, we assume the economy goes back to the original steady state as $t \to \infty$, and in practice, we truncate the horizon at some large $t = T$.

Given a candidate tightness path $\boldsymbol{\theta}$, we first solve the surplus equation (15) backward from the terminal steady state. Importantly, conditional on θ, the HJB equation does not depend on any other aggregate variables, including the employment distribution. This procedure gives the surplus function $S_t(z; \mathbf{A}, \boldsymbol{\theta})$ and the associated separation cutoff below which a match separates, $\underline{z}_t(\mathbf{A}, \boldsymbol{\theta})$, at each point in time. Note that by definition, $S_t(\underline{z}_t; \mathbf{A}, \boldsymbol{\theta}) = 0$.

Given a candidate guess of $\boldsymbol{\theta}$ and the cutoff $\underline{z}_t(\mathbf{A}, \boldsymbol{\theta})$ obtained in the previous step, we can iterate the Kolmogorov forward equation (18) forward to obtain the entire path of the employment distribution, which we denote as $n_t(z; \boldsymbol{\theta}, \mathbf{A})$. This procedure also gives the unemployment rate $u_t(\boldsymbol{\theta}, \mathbf{A}) = 1 - \int_{\underline{z}_t}^\infty n_t(z; \boldsymbol{\theta}, \mathbf{A})dz$. We can then back out meeting shares from (6), which we write as $f_t^u(\boldsymbol{\theta}, \mathbf{A})$ and $f_t(z; \boldsymbol{\theta}, \mathbf{A})$.

The conjectured tightness path must then be consistent with the free-entry condition

(17) at every date:

$$\kappa = (1-\gamma)q(\theta_t)\Bigg[f_t^u(\boldsymbol{\theta},\mathbf{A})\int_{\underline{z}_t}^{\infty} g(z)S_t(z;\mathbf{A},\boldsymbol{\theta})dz + \int_0^{\infty}\int_z^{\infty} f_t(z;\boldsymbol{\theta},\mathbf{A})g(z')\left[S_t(z';\mathbf{A},\boldsymbol{\theta}) - \omega S_t(z;\mathbf{A},\boldsymbol{\theta})\right]dz'dz\Bigg].$$

We solve this system nonlinearly by updating the candidate tightness path $\boldsymbol{\theta}$, repeating the backward value calculation and the forward employment calculation at each guess.

The planner's transition is obtained in the same way, using (21) for the match value $\mu_t(z)$ with endogenous separation, and (23) for vacancy creation.

G Wage Expression

We derive the expression for the wage in the steady state. Evaluating (10) at the steady state, the wage of a job with match quality z and outside option $\bar{O}$ is given by

$$\begin{aligned} w(z,\bar{O}) = Az - rJ(z,\bar{O}) &+ \zeta p(\theta)\int_z^{\infty} g(z')\left(V - J(z,\bar{O})\right)dz' \\ &+ \zeta p(\theta)\int_0^{z} g(z')\left(J(z,\max\{\omega S(z'),\bar{O}\}) - J(z,\bar{O})\right)dz' \\ &+ \sigma\left(V - J(z,\bar{O})\right). \end{aligned}$$

Substituting free entry $V = 0$ and the bargaining solution (14), we can simplify the above expression as

$$w(z,\bar{O}) = Az - (1-\gamma)\Bigg[(r+\sigma+\zeta p(\theta)(1-G(z)))\Big(S(z) - \bar{O}\Big) + \zeta p(\theta)\int_0^{z} g(z')\max\{\omega S(z') - \bar{O}, 0\}dz'\Bigg].$$

In order to study the wage distribution, we need to keep track of the joint distribution of the match quality and the outside option. Let $N(z, y)$ be the cumulative distribution function of the joint distribution of the match quality and the outside option below $S(y)$. Note that $N(z, \underline{z})$ corresponds to the density with match quality z with unemployment outside option $\bar{O} = 0$ and is a mass point.

The steady state $N(z, \underline{z})$ satisfies the following stock-flow equation:

$$0 = p(\theta)ug(z) - [\sigma + \zeta p(\theta)(1 - G(\underline{z}))] N(z, \underline{z}), \tag{64}$$

and $N(z, y)$ with $y \in (\underline{z}, z)$ satisfies

$$0 = \zeta p(\theta)g(z)n(y) + \zeta p(\theta)g(y)N(z, y) - [\sigma + \zeta p(\theta)(1 - G(y))] n(z, y), \tag{65}$$

where $n(z, y) \equiv \partial_y N(z, y)$ and $n(y)$ is

$$n(y) = \frac{p(\theta)ug(y)[\sigma + \zeta p(\theta)(1 - G(\underline{z}))]}{[\sigma + \zeta p(\theta)(1 - G(y))]^2}.$$

Solving for $N(z, \underline{z})$ from (64), we have

$$\begin{aligned} N(z, \underline{z}) &= \frac{p(\theta)ug(z)}{[\sigma + \zeta p(\theta)(1 - G(\underline{z}))]} \\ &= n(z) \left(\frac{\sigma + \zeta p(\theta)(1 - G(z))}{\sigma + \zeta p(\theta)(1 - G(\underline{z}))} \right)^2. \end{aligned}$$

We can rewrite (65) as

$$\partial_y \Big\{ [\sigma + \zeta p(\theta)(1 - G(y))] N(z, y) \Big\} = \zeta p(\theta)g(z)n(y),$$

which we can integrate as

$$\begin{aligned}[\sigma+\zeta p(\theta)(1-G(y))]\,N(z,y) &= \zeta p(\theta)g(z)\int_{\underline{z}}^{y} n(z')dz' + [\sigma+\zeta p(\theta)(1-G(\underline{z}))]\,N(z,\underline{z}) \\ &= \zeta p(\theta)g(z)\int_{\underline{z}}^{y} n(z')dz' + p(\theta)ug(z) \qquad (66)\end{aligned}$$

Note that, using $\frac{d}{dz'}\left[\sigma+\zeta p(\theta)(1-G(z'))\right] = -\zeta p(\theta)g(z')$,

$$\begin{aligned}\int_{\underline{z}}^{y} n(z')\,dz' &= \int_{\underline{z}}^{y} \frac{p(\theta)\,u\,g(z')\left[\sigma+\zeta p(\theta)(1-G(\underline{z}))\right]}{\left[\sigma+\zeta p(\theta)(1-G(z'))\right]^2}\,dz' \\ &= \frac{p(\theta)\,u\left[\sigma+\zeta p(\theta)(1-G(\underline{z}))\right]}{\zeta p(\theta)}\left[\frac{1}{\sigma+\zeta p(\theta)(1-G(y))} - \frac{1}{\sigma+\zeta p(\theta)(1-G(\underline{z}))}\right] \\ &= \frac{u}{\zeta}\left(\frac{\sigma+\zeta p(\theta)(1-G(\underline{z}))}{\sigma+\zeta p(\theta)(1-G(y))} - 1\right). \qquad (67)\end{aligned}$$

Substituting (67) into (66), we have

$$\begin{aligned}N(z,y) &= p(\theta)ug(z)\frac{\sigma+\zeta p(\theta)(1-G(\underline{z}))}{\left[\sigma+\zeta p(\theta)(1-G(y))\right]^2} \\ &= n(z)\frac{\left[\sigma+\zeta p(\theta)(1-G(z))\right]^2}{\left[\sigma+\zeta p(\theta)(1-G(y))\right]^2}.\end{aligned}$$

H No Reservation Match Quality Margin

In the baseline model, we assumed that the lower bound of the distribution of match quality is low enough so that some matches are not formed after meeting. In this section, we alternatively assume that the lower bound of the distribution of match quality is high enough so that all matches are formed after meeting.

Both the decentralized equilibrium and the planner's solution are defined in the same way as in the baseline model except that (16) and (22) are replaced with $\underline{z}_t = z_{\min}$, where $z_{\min}$ is the lower bound of the distribution of match quality. Because the cutoff is fixed, each steady-state allocation is pinned down by a single free-entry equation in the sin-

gle unknown θ. Excess vacancies follow because, at the planner's tightness under Hosios, private values dominate social values, so private free entry requires a higher θ. The excessive-vacancy result then follows from three steps: (i) at $\theta = \theta^{SP}$ under the Hosios condition, private match values dominate social ones; (ii) hence the private vacancy return exceeds κ; (iii) the private vacancy return is decreasing in θ, so $\theta^{DE} > \theta^{SP}$.

Steady-state decentralized equilibrium. Write $\underline{z} \equiv z_{\min}$ and $\Gamma(z;\theta) \equiv r + \sigma + (1 - \omega(1-\gamma))\zeta p(\theta)(1 - G(z))$ as in Lemma 1. The private surplus solves the same ODE as in the baseline,

$$[r + \sigma + (1 - \omega(1-\gamma))\zeta p(\theta)(1 - G(z))]\, S'(z) = A + (1-\gamma)(1-\omega)\zeta p(\theta) g(z) S(z),$$

but the boundary condition is no longer $S(\underline{z}) = 0$. Evaluating (15) at $z = \underline{z}$ in the steady state and solving the ODE yields

$$S(z;\theta) = S(\underline{z};\theta)\left(\frac{\Gamma(z;\theta)}{\Gamma(\underline{z};\theta)}\right)^{\frac{(\omega-1)(1-\gamma)}{1-\omega(1-\gamma)}} + A\int_{\underline{z}}^{z} \frac{\Gamma(z;\theta)^{\frac{(\omega-1)(1-\gamma)}{1-\omega(1-\gamma)}}}{\Gamma(\tilde{z};\theta)^{\frac{\gamma}{1-\omega(1-\gamma)}}}\, d\tilde{z}, \qquad z \geq \underline{z}, \tag{68}$$

with

$$S(\underline{z};\theta) = \frac{A\underline{z} - h - \dfrac{A(1-\zeta)}{\zeta}\displaystyle\int_{\underline{z}}^{\infty}\left[1 - \left(\frac{r+\sigma}{\Gamma(z;\theta)}\right)^{\frac{\gamma}{1-\omega(1-\gamma)}}\right] dz}{\Gamma(\underline{z};\theta)\left\{1 + \dfrac{1-\zeta}{\zeta}\left[1 - \left(\frac{r+\sigma}{\Gamma(\underline{z};\theta)}\right)^{\frac{\gamma}{1-\omega(1-\gamma)}}\right]\right\}}, \tag{69}$$

where we used $G(\underline{z}) = 0$. Market tightness is pinned down by free entry

$$\kappa = R^{DE}(\theta) \equiv (1-\gamma)q(\theta)\left[f^u \int_{\underline{z}}^{\infty} g(z) S(z;\theta)\, dz + \int_{\underline{z}}^{\infty}\int_{z}^{\infty} f(z) g(z') \left[S(z';\theta) - \omega S(z;\theta)\right] dz'\, dz\right], \tag{70}$$

where f^u and $f(z)$ are the steady-state meeting shares in (6) evaluated at $\underline{z} = z_{\min}$.

Steady-state planner's solution. Likewise, the planner's value solves

$$\big[r+\sigma+\zeta p(\theta)(1-G(z))\big]\mu'(z)=A,$$

with boundary condition obtained by evaluating (21) at $z=\underline{z}$. Substituting the planner's free-entry condition as in the proof of Lemma 2 and solving the ODE yields

$$\mu(\underline{z};\theta)=\frac{A\underline{z}-h-A(1-\zeta)p(\theta)\displaystyle\int_{\underline{z}}^{\infty}\frac{\sigma\eta(\theta)+\zeta p(\theta)(1-G(z))}{\sigma+\zeta p(\theta)(1-G(z))}\frac{1-G(z)}{r+\sigma+\zeta p(\theta)(1-G(z))}\,dz}{r+\sigma+p(\theta)-(1-\eta(\theta))(1-\zeta)p(\theta)\dfrac{\sigma}{\sigma+\zeta p(\theta)}},\tag{71}$$

and

$$\mu(z;\theta)=\mu(\underline{z};\theta)+A\int_{\underline{z}}^{z}\frac{1}{r+\sigma+\zeta p(\theta)(1-G(\tilde{z}))}\,d\tilde{z},\qquad z\geq\underline{z}.$$

The efficient tightness solves

$$\kappa=(1-\eta(\theta))q(\theta)\left[f^{u}\int_{\underline{z}}^{\infty}g(z)\mu(z;\theta)\,dz+\int_{\underline{z}}^{\infty}\int_{z}^{\infty}f(z)g(z')\big[\mu(z';\theta)-\mu(z;\theta)\big]\,dz'\,dz\right].\tag{72}$$

Equilibrium vs. planner's solution. The following lemma shows that, for a given value of θ, the equilibrium private surplus is larger than the planner's value.

Lemma 3 *Fix θ and assume $\gamma\leq\eta(\theta)$. Then, in the above environment,*

$$S(z;\theta)\geq\mu(z;\theta),\quad \textit{for all } z\geq\underline{z}.$$

Proof. Suppress the dependence of S and μ on θ when convenient. As in the proof of

Proposition 1,

$$\frac{1}{\zeta p(\theta)}\left[1-\left(\frac{r+\sigma}{\Gamma(z;\theta)}\right)^{\frac{\gamma}{1-\omega(1-\gamma)}}\right] \leq \frac{\gamma(1-G(z))}{r+\sigma+\gamma\zeta p(\theta)(1-G(z))}$$
$$\leq \frac{\sigma\eta(\theta)+\zeta p(\theta)(1-G(z))}{\sigma+\zeta p(\theta)(1-G(z))}\frac{1-G(z)}{r+\sigma+\zeta p(\theta)(1-G(z))},$$

where the second inequality uses $\gamma \leq \eta(\theta)$. Thus, the numerator in (69) is weakly larger than that in (71).

Next we compare the denominators. The denominator in (69) is

$$D^{DE} \equiv \{r+\sigma+(1-\omega(1-\gamma))\zeta p(\theta)\}\left\{1+\frac{1-\zeta}{\zeta}\left[1-\left(\frac{r+\sigma}{r+\sigma+(1-\omega(1-\gamma))\zeta p(\theta)}\right)^{\frac{\gamma}{1-\omega(1-\gamma)}}\right]\right\}$$

$$\leq \{r+\sigma+(1-\omega(1-\gamma))\zeta p(\theta)\}\left\{1+\frac{1-\zeta}{\zeta}\left[1-\left(\frac{r+\sigma}{r+\sigma+\gamma\zeta p(\theta)}\right)\right]\right\}$$
$$= \{r+\sigma+(1-\omega(1-\gamma))\zeta p(\theta)\}\left\{1+\frac{\gamma(1-\zeta)p(\theta)}{r+\sigma+\gamma\zeta p(\theta)}\right\}$$
$$\leq \big(r+\sigma+\zeta p(\theta)\big)\left\{1+\frac{\gamma(1-\zeta)p(\theta)}{r+\sigma+\gamma\zeta p(\theta)}\right\}$$
$$= \big(r+\sigma+\zeta p(\theta)\big)+(1-\zeta)p(\theta)\frac{\gamma\big(r+\sigma+\zeta p(\theta)\big)}{r+\sigma+\gamma\zeta p(\theta)}$$
$$\leq \big(r+\sigma+\zeta p(\theta)\big)+(1-\zeta)p(\theta)\frac{\sigma\gamma+\zeta p(\theta)}{\sigma+\zeta p(\theta)}$$
$$\leq \big(r+\sigma+\zeta p(\theta)\big)+(1-\zeta)p(\theta)\frac{\sigma\eta(\theta)+\zeta p(\theta)}{\sigma+\zeta p(\theta)} \equiv D^{SP},$$

where D^{SP} is the denominator in (71), and the second inequality uses the same inequality

as in the proof of Proposition 1. The second-to-last inequality follows because

$$\begin{aligned}
\frac{\gamma\big(r+\sigma+\zeta p(\theta)\big)}{r+\sigma+\gamma\zeta p(\theta)} &\leq \frac{\sigma\gamma+\zeta p(\theta)}{\sigma+\zeta p(\theta)} \\
\gamma\big(r+\sigma+\zeta p(\theta)\big)\big(\sigma+\zeta p(\theta)\big) &\leq \big(\sigma\gamma+\zeta p(\theta)\big)\big(r+\sigma+\gamma\zeta p(\theta)\big) \\
\gamma r\sigma+\gamma\sigma^2+\gamma r\zeta p(\theta)+2\gamma\sigma\zeta p(\theta)+\gamma\big(\zeta p(\theta)\big)^2 &\leq \gamma r\sigma+\gamma\sigma^2+\gamma^2\sigma\zeta p(\theta)+r\zeta p(\theta)+\sigma\zeta p(\theta)+\gamma\big(\zeta p(\theta)\big)
\end{aligned}$$

$$\begin{aligned}
\gamma r\zeta p(\theta)+2\gamma\sigma\zeta p(\theta) &\leq \gamma^2\sigma\zeta p(\theta)+r\zeta p(\theta)+\sigma\zeta p(\theta) \\
\gamma r+2\gamma\sigma &\leq \gamma^2\sigma+r+\sigma \\
0 &\leq (1-\gamma)\big[r+\sigma(1-\gamma)\big].
\end{aligned}$$

The last inequality uses $\gamma \leq \eta(\theta)$. Since the numerator is larger and the denominator is smaller, $S(\underline{z}) \geq \mu(\underline{z})$. Because all draws are accepted, $\mu(\underline{z}) \geq 0$, and hence $S(\underline{z}) \geq 0$. Comparing the two ODEs,

$$S'(z) = \frac{A+(1-\gamma)(1-\omega)\zeta p(\theta)g(z)S(z)}{\Gamma(z;\theta)} \geq \frac{A}{r+\sigma+\zeta p(\theta)(1-G(z))} = \mu'(z).$$

Consequently, $S(z) \geq \mu(z)$ for all $z \geq \underline{z}$. ■

Lemma 4 *For any $\bar{\theta}$ such that $S(\underline{z};\bar{\theta}) \geq 0$, $R^{DE}(\theta)$ is strictly decreasing in θ on $(0,\bar{\theta}]$.*

Proof. Multiplying (69) by $\Gamma(\underline{z};\theta)^{\frac{(1-\omega)(1-\gamma)}{1-\omega(1-\gamma)}}$ leaves the numerator unchanged and yields

$$\begin{aligned}
&S(\underline{z};\theta)\,\Gamma(\underline{z};\theta)^{\frac{(1-\omega)(1-\gamma)}{1-\omega(1-\gamma)}} \\
&= \frac{A\underline{z}-h-\dfrac{A(1-\zeta)}{\zeta}\displaystyle\int_{\underline{z}}^{\infty}\left[1-\left(\frac{r+\sigma}{\Gamma(z;\theta)}\right)^{\frac{\gamma}{1-\omega(1-\gamma)}}\right]dz}{\Gamma(\underline{z};\theta)^{\frac{\gamma}{1-\omega(1-\gamma)}}+\dfrac{1-\zeta}{\zeta}\left[\Gamma(\underline{z};\theta)^{\frac{\gamma}{1-\omega(1-\gamma)}}-(r+\sigma)^{\frac{\gamma}{1-\omega(1-\gamma)}}\right]},
\end{aligned}$$

which is decreasing in θ. Therefore, since both the first and the second terms in (68) are decreasing in θ, $S(z;\theta)$ is weakly decreasing in θ for each $z \geq \underline{z}$.

As in the proof of Lemma 1, (70) can be rewritten as

$$R^{DE}(\theta) = (1-\gamma)q(\theta)\int_{\underline{z}}^{\infty} \frac{\sigma g(z)\left[\sigma + (1-\omega)\zeta p(\theta)(1-G(z))\right]}{\left[\sigma + \zeta p(\theta)(1-G(z))\right]^2} S(z;\theta)\, dz.$$

The weight on $S(z;\theta)$ is nonnegative and weakly decreasing in θ. Since $q'(\theta) < 0$, R^{DE} is strictly decreasing on $(0, \bar{\theta}]$ wherever it is positive. ■

Excessive market tightness. We are now ready to state the main result, which extends Proposition 1 to the case where the reservation match quality margin is absent.

Proposition 8 *Suppose $\gamma = \eta(\theta^{SP})$. Then, in the environment described above, the steady-state equilibrium market tightness is higher than the socially efficient level when $\zeta > 0$.*

Proof. Fix $\theta = \theta^{SP}$. Lemma 3 yields $S(z;\theta^{SP}) \geq \mu(z;\theta^{SP})$ for all $z \geq \underline{z}$, and in particular $S(\underline{z};\theta^{SP}) \geq 0$. The ODE comparison in that lemma also gives $\partial_z S(z;\theta^{SP}) \geq \partial_z \mu(z;\theta^{SP})$, so for $z' > z$,

$$S(z') - \omega S(z) = S(z') - S(z) + (1-\omega)S(z) \geq \mu(z') - \mu(z).$$

At this common tightness, the two allocations have the same f^u and $f(z)$, and Hosios equates the prefactors in (70) and (72). Combined with $S(z) \geq \mu(z)$ and $S(z') - \omega S(z) \geq \mu(z') - \mu(z)$ for $z' > z$, this implies $R^{DE}(\theta^{SP}) \geq \kappa$.

Strictness follows by distinguishing two cases. If $\omega < 1$, the worker-stealing term $(1-\omega)\int_{\underline{z}}^{\infty} f(z)[1-G(z)]S(z)\, dz$ is strictly positive: $S(z) \geq \mu(z) > 0$ for $z > \underline{z}$, and $f(z)[1-G(z)] > 0$ on a set of positive measure. If $\omega = 1$, the ODE comparison gives $S'(z) > \mu'(z)$ whenever $1 - G(z) > 0$, since $\gamma < 1$ and $\zeta > 0$. Together with $S(\underline{z}) \geq \mu(\underline{z})$ and $f^u > 0$, this implies $f^u \int_{\underline{z}}^{\infty} g(z)[S(z) - \mu(z)]\, dz > 0$. Thus, $R^{DE}(\theta^{SP}) > \kappa$ in either case.

By Lemma 4, R^{DE} is strictly decreasing on $(0, \theta^{SP}]$. Hence $R^{DE}(\theta) > \kappa$ for all $\theta \leq \theta^{SP}$, and $R^{DE}(\theta^{DE}) = \kappa$ implies $\theta^{DE} > \theta^{SP}$. ■

	Market tightness		Reservation match quality		Consumption		Unemployment rate (%)	
	DE	SP	DE	SP	DE	SP	DE	SP
0. Baseline	1.00	0.78	1.221	1.228	1.4624	1.4633	5.5	6.4
1. $h/\mathbb{E}[w] = 0.6$	1.00	0.77	0.876	0.879	0.9830	0.9835	5.5	6.4
2. No reservation margin ($h = 0$)	1.00	0.75	0.400	0.400	0.4586	0.4593	5.5	6.3
3. Log-normal match quality	1.00	0.69	1.229	1.236	1.4533	1.4563	5.5	6.5

Table I.4: Steady-state robustness summary: decentralized equilibrium vs. efficient allocation

I Alternative Parameterization

We explore the robustness of our quantitative exercise under three alternative parameterizations.

Table I.4 summarizes the steady-state decentralized and efficient allocations under the baseline and the three robustness exercises below. Across specifications, equilibrium market tightness remains above the planner's, and the unemployment rate remains lower.

I.1 Higher replacement rate

In the baseline calibration, we set the replacement rate to $h/\mathbb{E}[w] = 0.40$. We recalibrate the Pareto shape α to target $h/\mathbb{E}[w] = 0.60$.

Row 1 of Table I.4 reports the resulting steady-state comparison. The results are overall similar to the baseline calibration.

I.2 No Reservation Match Quality Margin

We explore an alternative parameterization where all meetings form matches, so that the reservation match quality margin is inactive.

Table I.5 shows the calibration. We set $h = 0$ and keep the Pareto shape α at the baseline value. We then choose $(\sigma, \kappa, m, \zeta)$ to match the same targets as in the baseline calibration except for $h/\mathbb{E}[w]$. We confirm that, under this parameterization, all meetings form matches.

Parameter	Description	Value	Target/Source
PANEL A. ASSIGNED PARAMETERS			
r	Discount rate	0.004	Annual interest rate 5%
η	Elasticity of matching function	0.5	Standard
γ	Worker bargaining power	0.5	Standard
ω	Offer matching	1.0	Dey and Flinn (2005), Cahuc et al. (2006)
h	Home production value	0	Shut down reservation margin
A	Aggregate productivity	1.0	Normalization
α	Pareto shape	7.49	Baseline value
$z_{\min}$	Pareto scale	0.40	Normalization
PANEL B. CALIBRATED PARAMETERS			
σ	Separation rate	0.024	EU rate 2.4%
κ	Cost of vacancy creation	0.16	Market tightness 1
m	Matching efficiency	0.41	Unemployment rate 5.5%
ζ	On-the-job search efficiency	0.24	EE rate 2.5%

Table I.5: Parameter values with no active reservation margin ($h = 0$)

Row 2 of Table I.4 reports the resulting steady-state comparison. The results are overall similar to the baseline calibration.

I.3 Log-normal match quality distribution

In the quantitative exercise of the baseline model, we assumed that the distribution of match quality is Pareto. An alternative common parametric assumption is the log-normal distribution. Here, we instead assume that the distribution of match quality is log-normal.

Table I.6 shows our calibration. We assume $\ln z$ follows a normal distribution with mean μ_N and variance σ_N^2. We calibrate $\{\mu_N, \sigma_N, m, \kappa, \zeta\}$ to the same targets as in the baseline, with one additional target of an average job-to-job wage gain of 8% (Barlevy, 2002).

Row 3 of Table I.4 reports the resulting steady-state comparison. The results are overall similar to the baseline calibration.

Parameter	Description	Value	Target/Source
PANEL A. ASSIGNED PARAMETERS			
r	Discount rate	0.004	Annual interest rate 5%
η	Elasticity of matching function	0.5	Standard
γ	Worker bargaining power	0.5	Standard
ω	Offer matching	1.0	Dey and Flinn (2005), Cahuc et al. (2006)
h	Home production value	0.6	Normalization
A	Aggregate productivity	1.0	Normalization
PANEL B. CALIBRATED PARAMETERS			
σ	Separation rate	0.024	EU rate 2.4%
μ_N	Mean of log-normal distribution	0.42	$h/\mathbb{E}[w] = 0.4$
σ_N	Std. of log-normal distribution	0.034	Avg. job-to-job wage gain 8%
κ	Cost of vacancy creation	0.20	Market tightness 1
m	Matching efficiency	0.41	Unemployment rate 5.5%
ζ	On-the-job search efficiency	0.24	EE rate 2.5%

Table I.6: Parameter values under log-normal match quality distribution

J Extension with endogenous effort in maintaining a match

Consider an extension of the baseline model with taxes to an environment with endogenous effort choice that determines the separation rate, similar to the setup in Balke and Lamadon (2022). In particular, we assume workers can choose the level of effort, e, which determines the separation rate of a match, $\sigma(e)$. The separation rate is decreasing and strictly convex in effort. The unit cost of effort is ι. The equilibrium joint match surplus in this environment is given by

$$\begin{aligned} rS_t(z) = \max_e\ & A_t z(1-\tau_t(z)) - h - \iota e - \sigma(e) S_t(z) \\ & + \zeta p(\theta_t) \int_z^\infty g(z') \left(\omega S_t(z) + \gamma \left[S_t(z') - \omega S_t(z)\right] - S_t(z)\right) dz' \\ & - p(\theta_t) \int_{\underline{z}_t}^\infty g(z') \gamma S_t(z') dz' + \dot{S}_t(z). \end{aligned}$$

An interior optimal effort choice for a job with match quality z satisfies

$$\iota = -\sigma'(e) S_t(z).$$

We denote the solution to the above expression as $e_t(z)$. Job-specific separation terms use $\sigma(e_t(z))$. In the employment-density equation (18), the separation outflow $\sigma n_t(z)$ is replaced by $\sigma(e_t(z))n_t(z)$. This replacement is the only modification to the decentralized equilibrium setup.

Likewise, the valuation of the job in the planner's solution is given by

$$\begin{aligned} r\mu_t(z) = \max_e\, & A_t z - h - \iota e - \sigma(e)\mu_t(z) \\ & - \int_{\underline{z}_t}^{\infty} p(\theta_t) g(z') \mu_t(z') dz' + \zeta p(\theta_t) \int_z^{\infty} g(z')(\mu_t(z') - \mu_t(z)) dz' \\ & + \kappa\theta_t(1-\zeta) + \dot{\mu}_t(z). \end{aligned}$$

An interior optimal effort choice for a job with match quality z satisfies

$$\iota = -\sigma'(e)\mu_t(z).$$

We denote the solution to the above expression as $e_t^{SP}(z)$. The planner's distribution and unemployment equations use the same aggregation with $e_t^{SP}(z)$.

Because private and social effort choices depend on $S_t(z)$ and $\mu_t(z)$ respectively, any tax that equates S and μ also equates effort. Instruments that do not close the valuation wedge leave effort distorted. The output tax $\tau(z)$ that ensures $\mu_t(z) = S_t(z)$ also ensures that effort choices are efficient, $e_t(z) = e_t^{SP}(z)$. By contrast, because unemployment insurance or a vacancy tax alone does not ensure $\mu_t(z) = S_t(z)$, effort choices will not be efficient.

K Additional Figures

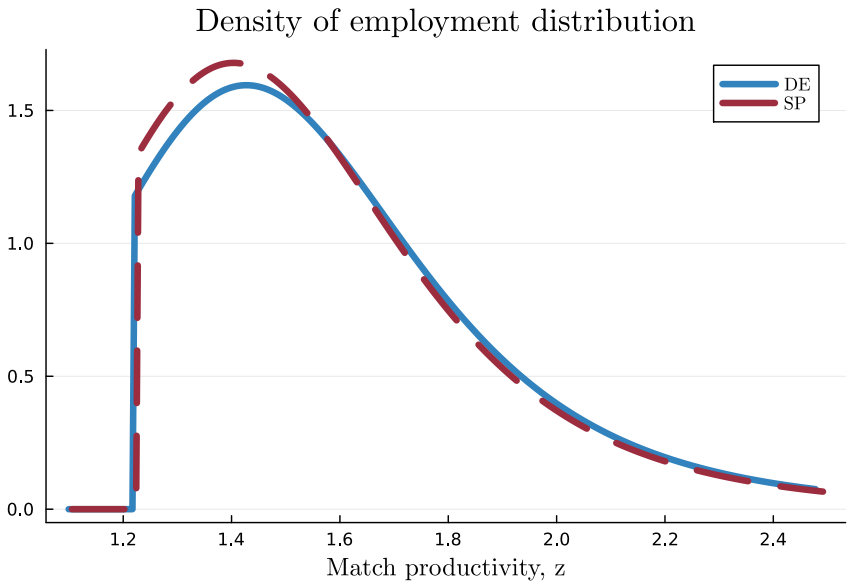


Figure K.4: Employment density: equilibrium vs. planner

Note: The figure plots the density of employment across match quality, $n(z)$, in the decentralized equilibrium (DE) and the efficient allocation (SP).

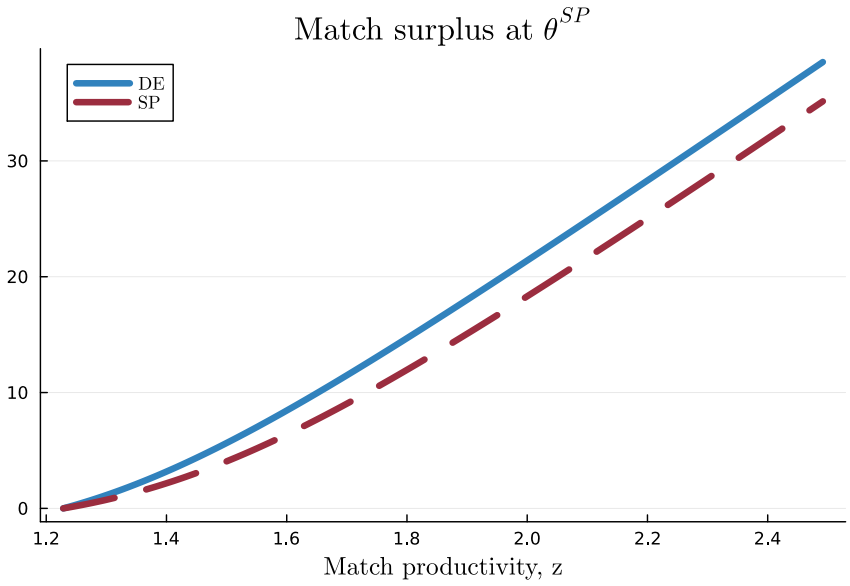


Figure K.5: Match surplus at the planner's tightness

Note: The figure plots the private match surplus $S(z)$ and the planner's valuation $\mu(z)$, both evaluated at the planner's steady-state tightness θ^{SP} and reservation match quality $\underline{z}^{SP}$, consistent with the valuation-wedge comparison.